Reinforcement Learning: Course Introduction

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Outline

1. Intelligence & RL
2. People: TA, Professor and His Academic Family
3. Course Information
4. Conclusion

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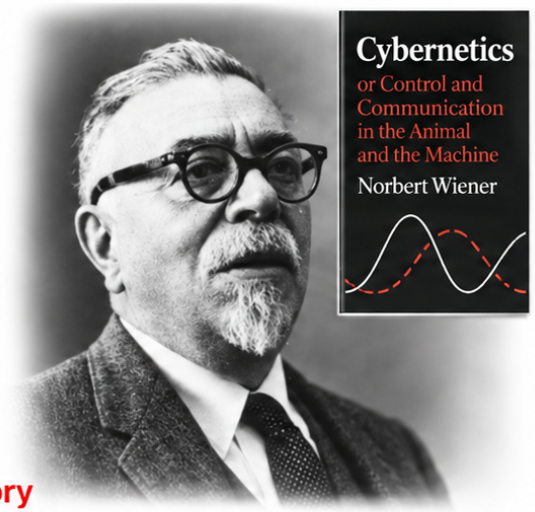
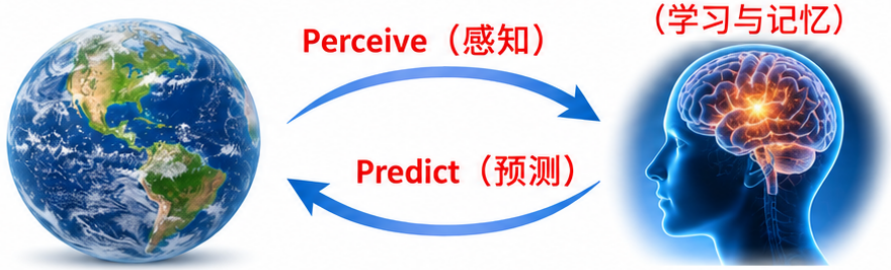
The Origin of Machine Intelligence

A More Magical Decade: the True Origin of Intelligence Study
Intelligence真正神奇的十年：智能研究的真正起源

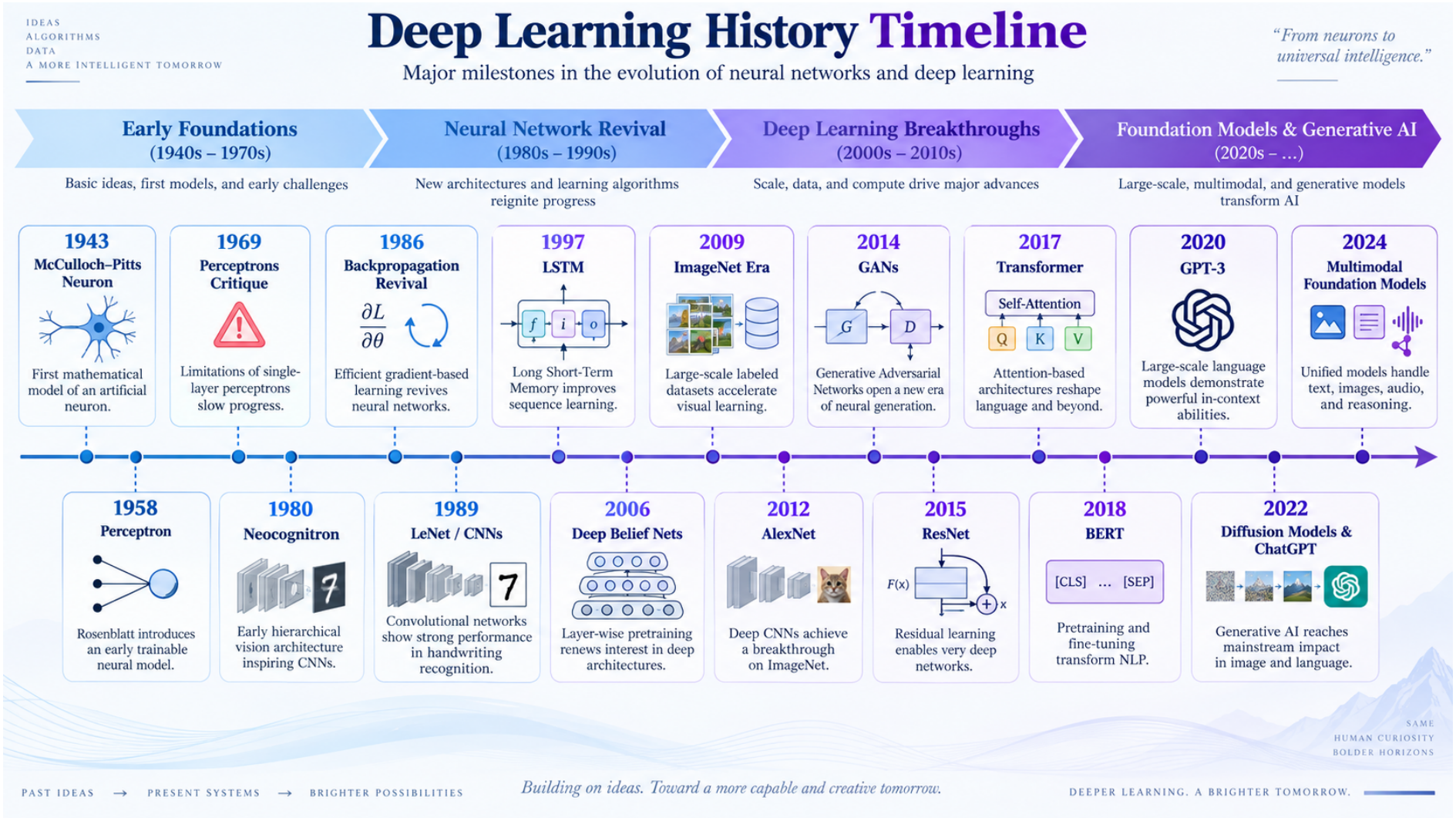
- 1943, **Artificial Neural Networks,** Warren McCulloch and Walter Pitts
  (1943年，**人工神经网络**，沃伦·麦卡洛克和沃尔特·皮茨)
- 1948, **Information Theory,** Claude Shannon
  (1948年，**信息论**，克劳德·香农)
- 1948, **Feedback Control & Cybernetics, Norbert Wiener**
  (1948年，**反馈控制与控制论**，**诺伯特·维纳**)
- 1944, **Game Theory,** John von Neumann
  (1944年，**博弈论**，约翰·冯·诺依曼)
- 1940's, **Turing Machine and Turing Test,** Alan Turing etc.
  (1940年代，**图灵机与图灵测试**，艾伦·图灵等。)



Norbert Wiener
(1894 – 1964)



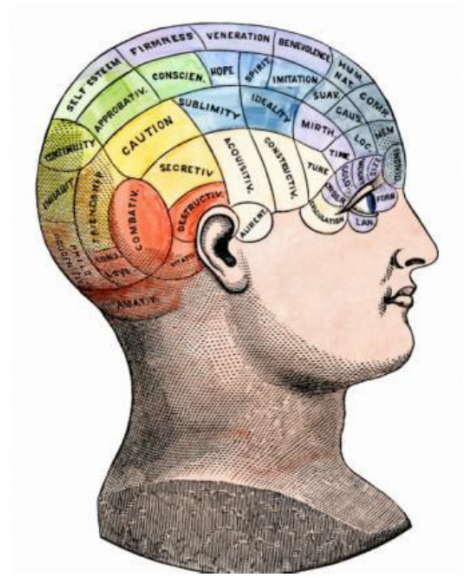
History Timeline of Machine Intelligence

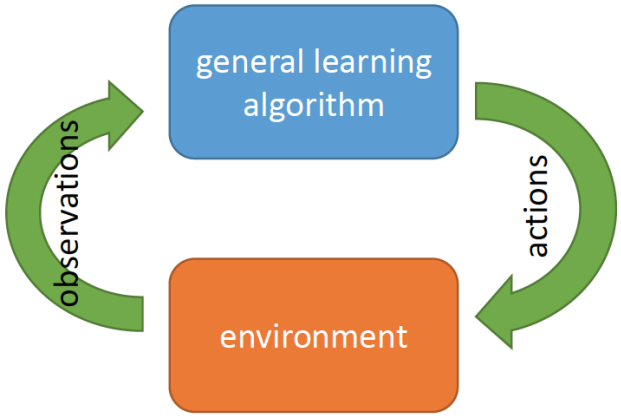


Our Focus: Interactive Intelligence

Instead of trying to produce a program to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain.

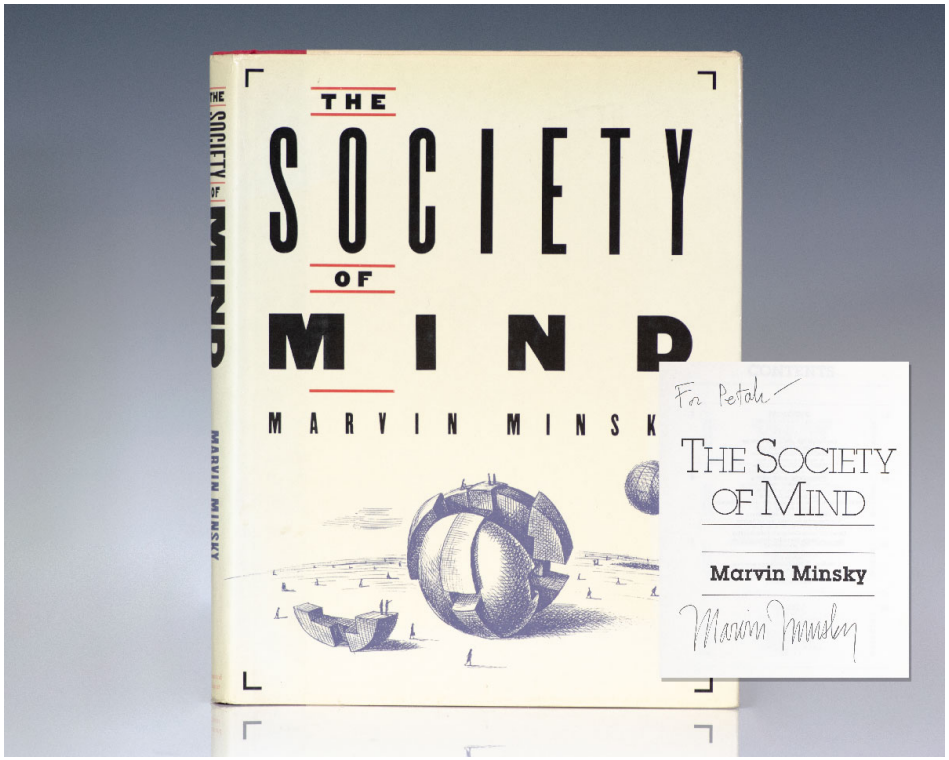
- Alan Turing





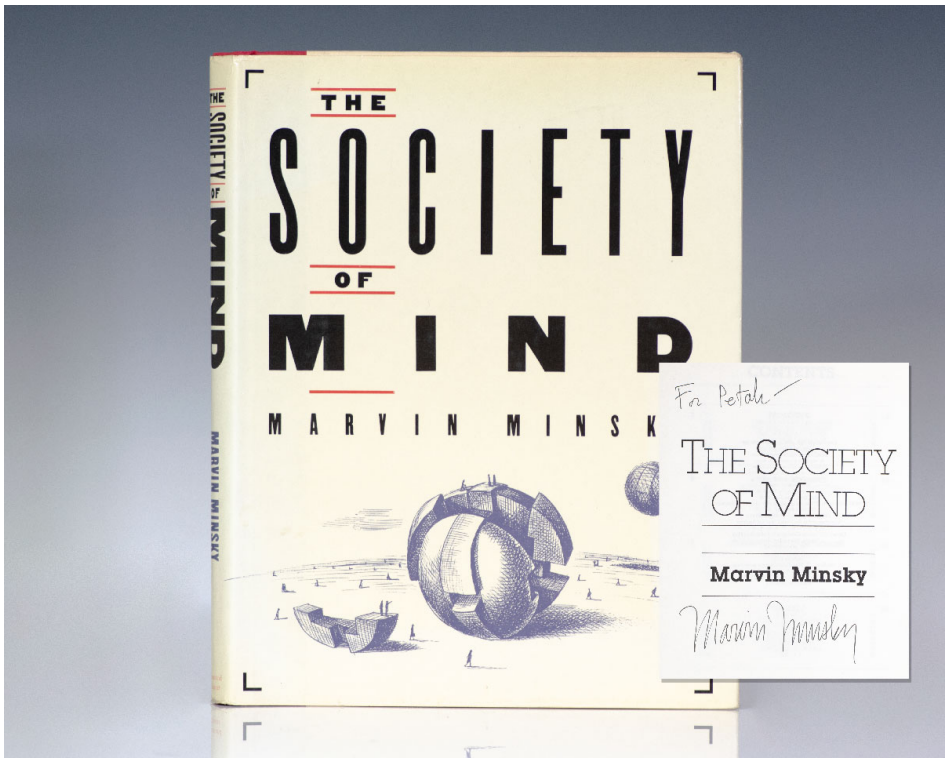
Intelligence through Agentic Interaction at Scale

- Marvin Minsky published a classical book "The Society of Mind" in 1986: Human intelligence was built step by step from the interactions of agents



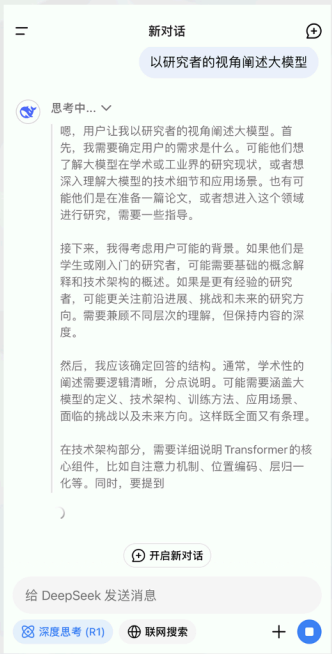
Intelligence through Agentic Interaction at Scale

- Famous quote: "What magical trick makes us intelligent? The trick is that there is no trick. The power of intelligence stems from our vast diversity, not from any single, perfect principle."

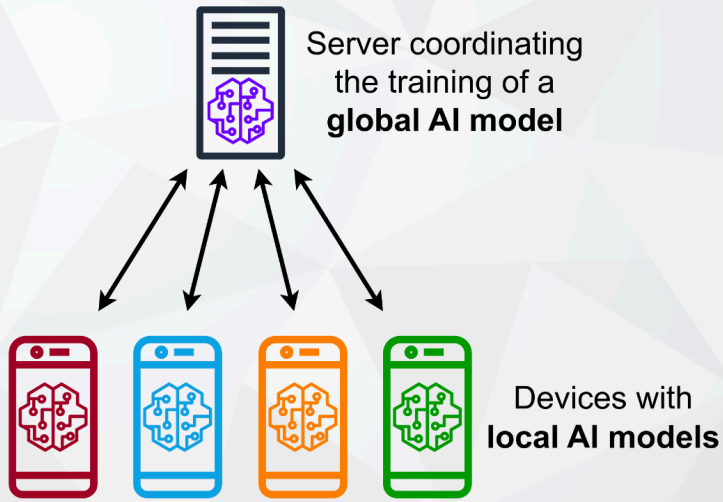


Example of Interactive Intelligence

Agent(人)与Agent(大模型)交互

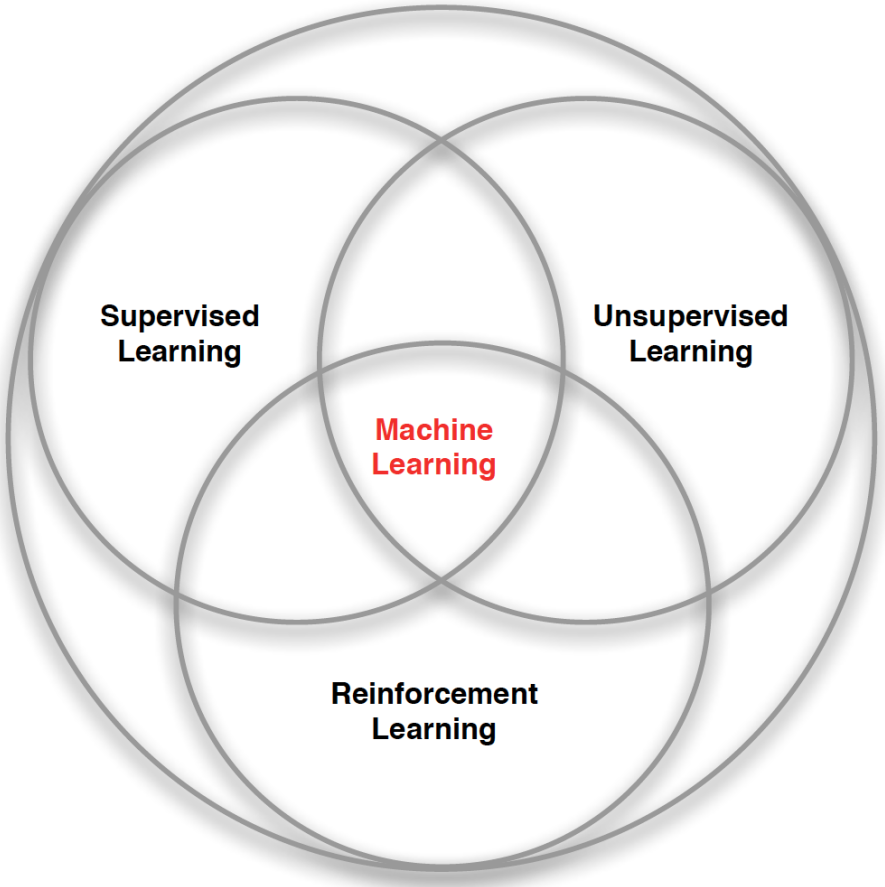


Agent(边缘服务器)与Agent(终端设备)交互



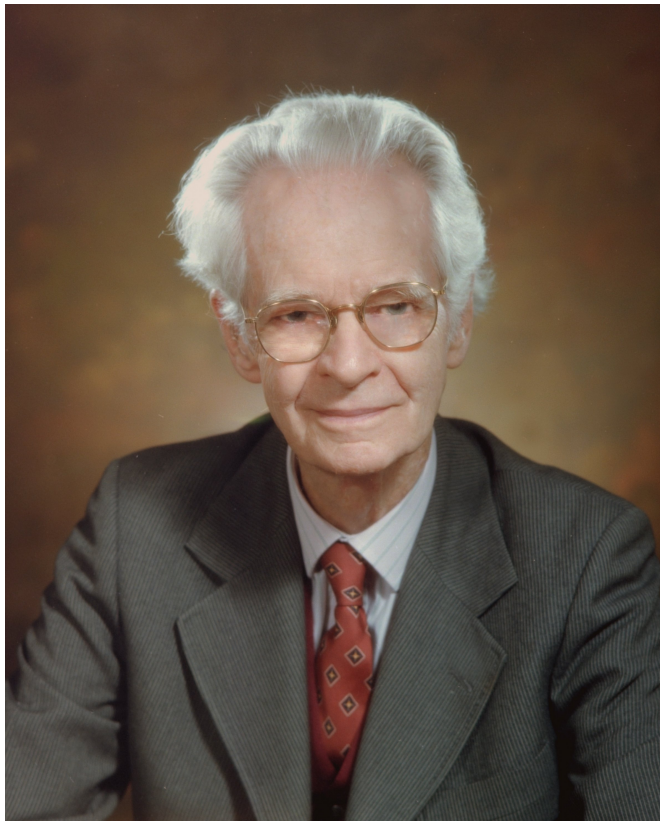
Modeling Tool of Interactive Intelligence

- Reinforcement learning is an area of machine learning inspired by **behavioral psychology**, concerned with how **agents** ought to take **actions** in an environment so as to maximize some notion of cumulative **reward**.



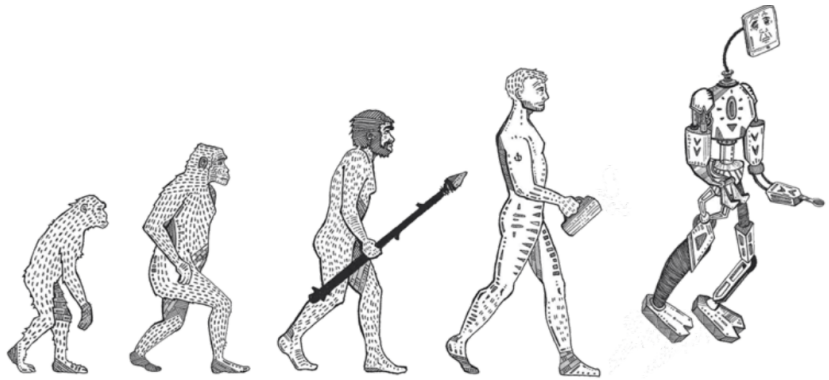
Behavioral Psychology

- Behavior is primarily shaped by **reinforcement** rather than free-will
  - behaviors that result in praise/pleasure tend to repeat
  - behaviors that result in punishment/pain tend to become extinct



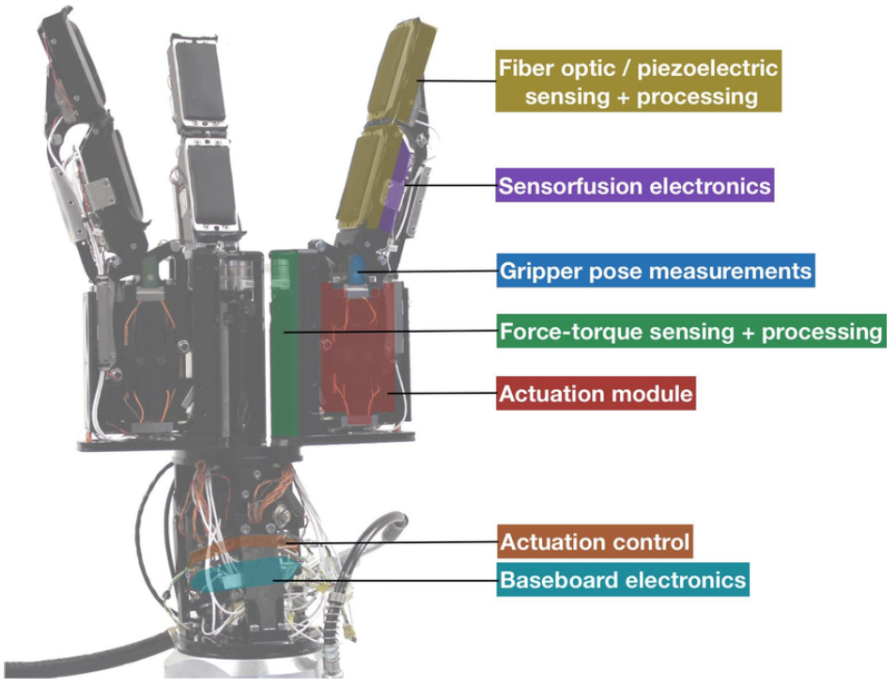
Agent

- An entity (learner & decision maker) that is equipped with
  - sensors, in order to sense the environment
  - end-effectors in order to act in the environment
  - goals that she wants to achieve



Action

- Used by the agent to interact with the environment.
- May have many different temporal granularities and abstractions



Actions can be

- The instantaneous torques(扭矩) applied on the gripper(夹爪)
- The instantaneous gripper translation, rotation, opening
- Instantaneous forces applied to the objects
- Short sequences of the above

Reward: Important Concept

- A **reward** R_t is a scalar feedback signal
- Indicates how well agent is doing at step t
- The agent's job is to maximize cumulative reward

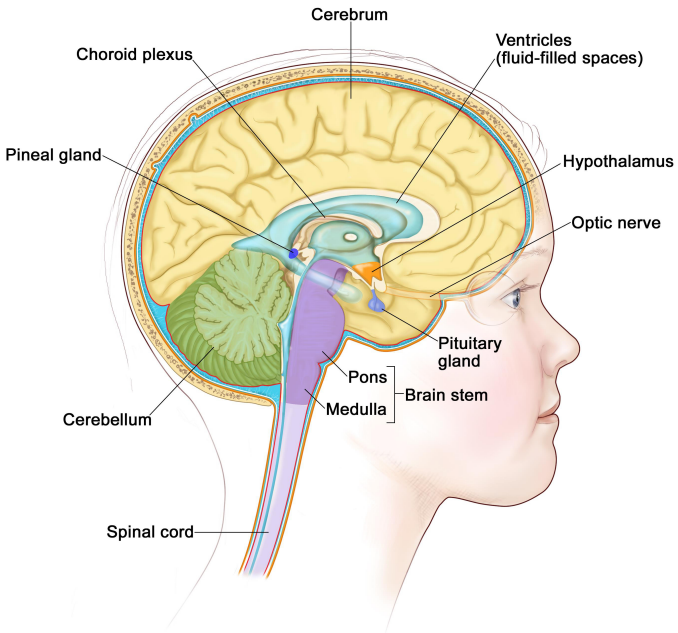
Reinforcement learning is based on the reward hypothesis:

Theorem (Von Neumann–Morgenstern Expected Utility Theorem)

All goals can be described by the maximization of expected cumulative reward

Reward: Brain Perspective

- Dopamine(多巴胺) & 内啡肽 (Endorphins)

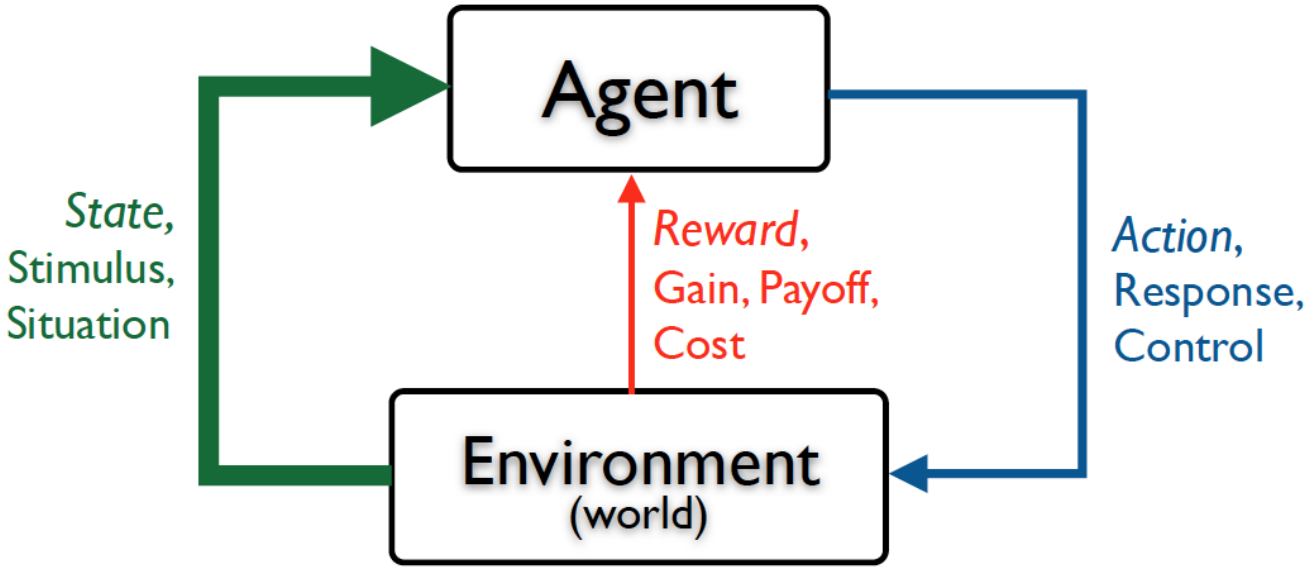


维度	多巴胺	内啡肽
主要作用	驱动欲望、即时奖励	缓解疼痛、长期满足
触发条件	预期奖励、新鲜刺激	身体压力、疼痛或耐力活动
持续时间	短暂（易消退）	较持久
成瘾风险	高（如游戏、药物成瘾）	低（健康途径触发）
情绪类型	兴奋、渴望	平静、幸福感

Reward: Animal Psychology

- Negative reinforcements: pain & hunger
- Positive reinforcements: pleasure & food
- Reinforcements used to train animals

Inside Reinforcement Learning



- Environment may be unknown, nonlinear, stochastic and complex
- Agent learns a mapping from states to actions: seeking to maximize its cumulative reward in the long run

Inside Reinforcement Learning?

- Agent-oriented learning:
  - explicitly consider the whole problem of goal-directed agent interacting with an uncertain environment
  - more realistic and ambitious than other kinds of machine learning
- Learning by trial and error, with only delayed evaluative feedback (reward)
  - the kind of machine learning most like nature learning
  - learning that can tell for itself when it is right or wrong
- Trade-off between exploration & exploitation

Other Elements of Reinforcement Learning

- **Environment**: the thing agent interacts with, comprising everything outside the agent
- **State**: captures whatever information is available to the agent about its environment
- **Policy**: a mapping from perceived states of the environment to actions to be taken when in those states
- **Value function**: the value of a state is the total amount of reward an agent can expect to accumulate over the future, starting from that state
- **Efficiently estimating values**: core of almost all reinforcement learning algorithms (Immediate reward vs. long-term value)
- **A model of the environment** (optional)

Reinforcement Learning: Examples

- Helicopter control
- Game playing (go, atari, backgammon)
- Robotics
- Autonomous vehicles

Example: Helicopter Control

- Agent: controller
- Environment: helicopter
- State: position, orientation, velocity & angular velocity
- Action: collective pitch(主旋翼升降), cyclic pitch(周期变距方向) & tail rotor control(尾桨控制)
- Reward: $-$deviation from desired trajectory
- 2008(Andrew Ng): automated helicopter wins acrobatic competition against humans

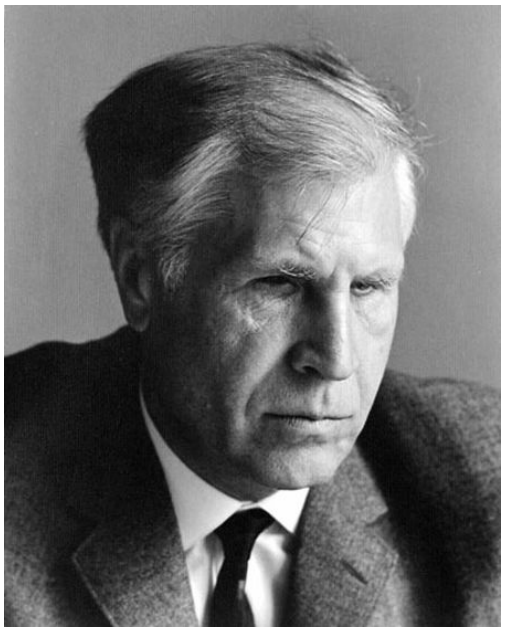
Characteristics of Reinforcement Learning

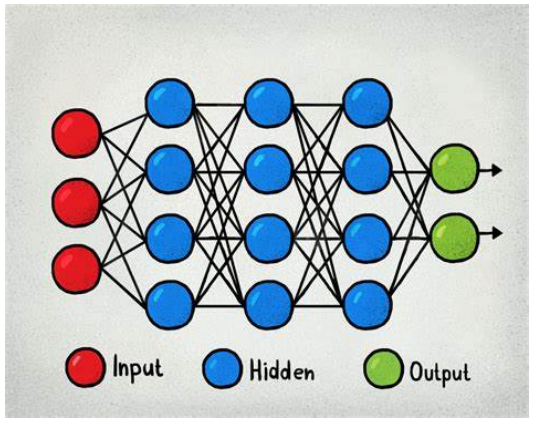
What makes reinforcement learning different from other machine learning paradigms?

- Multi-stage Learning with trial-and-error
- There is no supervisor, only a reward signal
- Feedback is delayed, not instantaneous
- Time really matters (sequential, non i.i.d data)
- Agent's actions affect the subsequent data it receives
- Trade-off between exploration & exploitation

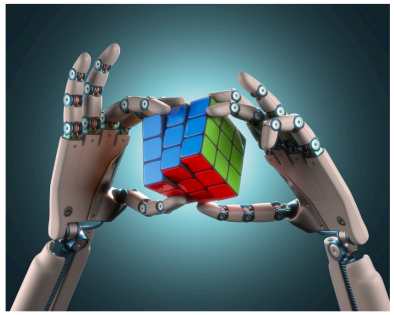
Reinforcement Learning: Interdisciplinary Topic

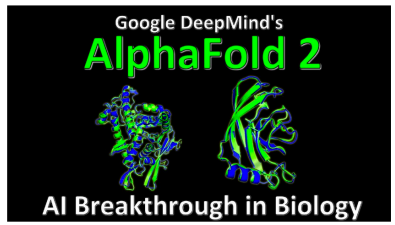
- Under Different Contexts, Reinforcement learning is also known as
  - Approximate optimal control
  - Approximate dynamic programming
  - Neuro-dynamic programming





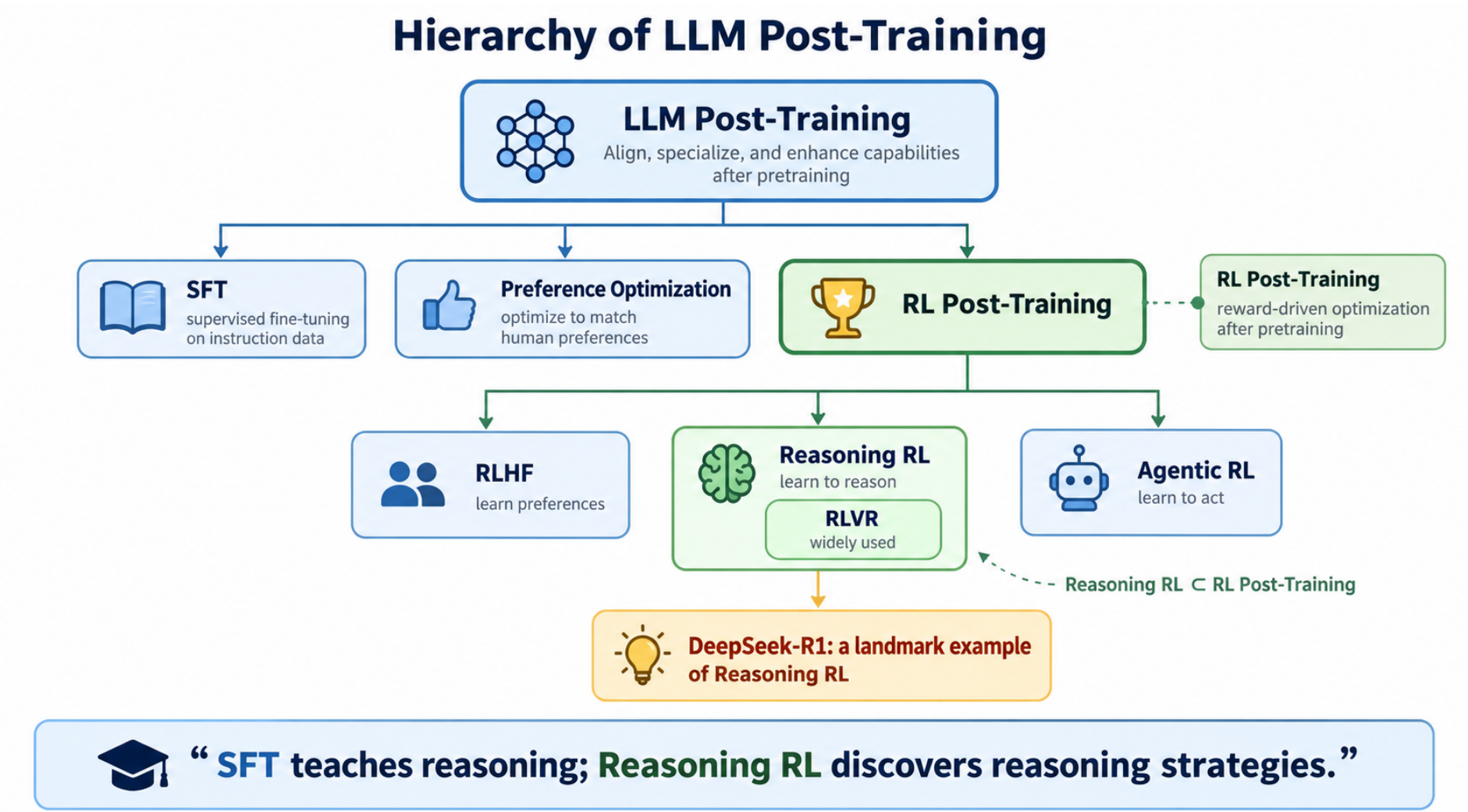
Successes in Reinforcement Learning





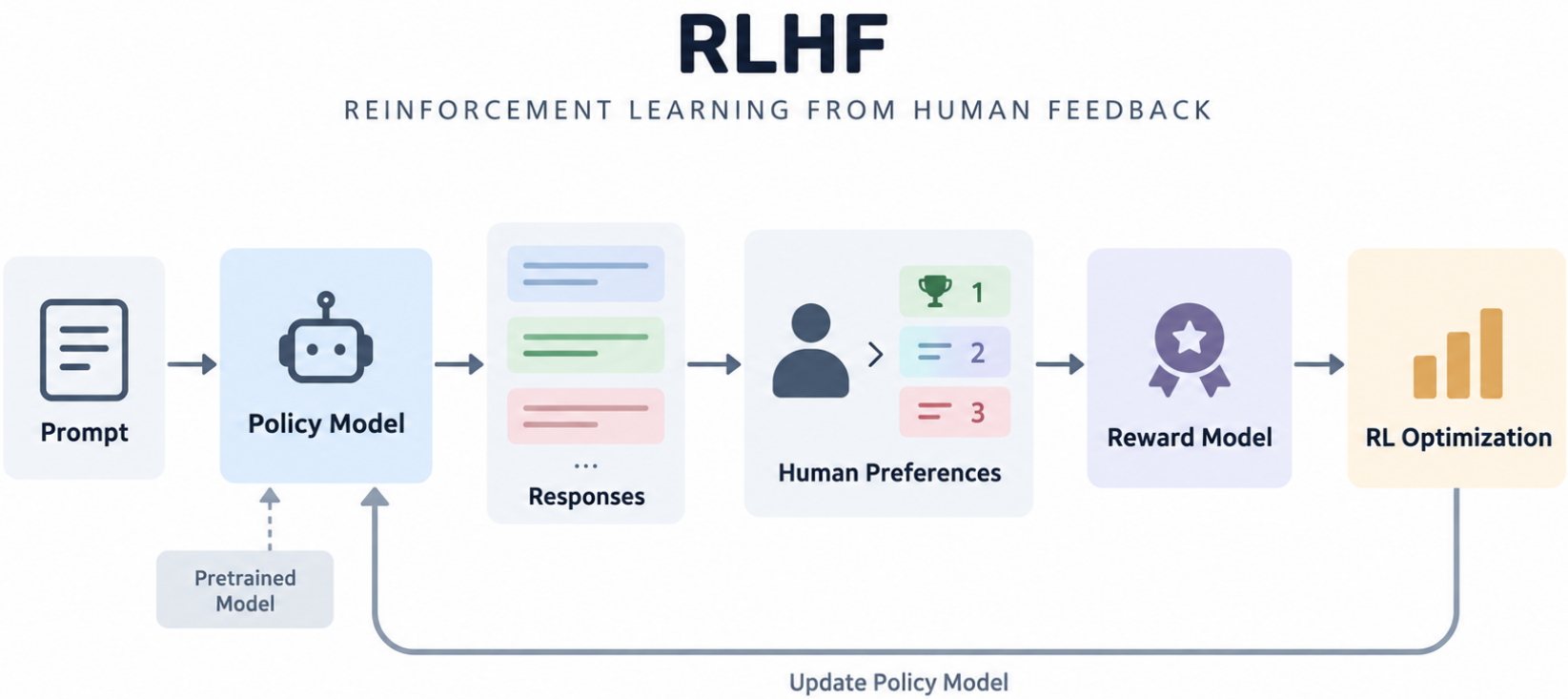
RL holds great promise in the era of AI

Recent Successes in Reinforcement Learning



Recent Successes in RL: RL Post-Training

Example of RL Post-Training: RLHF



Application: End-to-End RL with Chain of Thought

请用史记的风格来一段人工智能简史

Thought for 29 seconds

好的，用户之前让我用鲁迅、梁启超《少年中国说》和《滕王阁序》的风格写人工智能简史，现在又要求用《史记》的风格。我需要先回顾一下之前的回应，确保风格和结构符合用户期望。

首先，用户可能对历史或文学感兴趣，希望以不同文风呈现科技主题。这次是《史记》，属于纪传体通史，结构严谨，分本纪、列传等，语言古朴简练。我需要模仿《史记》的体例，比如用"太史公曰"结尾，引用人物言论，分章节等。

接下来，要确定人工智能发展的关键事件和人物，对应到《史记》的体例中。例如，图灵可立传，达特茅斯会议为本纪，神经网络的发展为列传等。注意用词要符合古文风格，如"世家"、"列传"等。

同时，用户之前的例子用了很多比喻和典故，如鲁迅中的孔乙己、祥林嫂，梁启超中的比喻等。在《史记》风格中，可能需要更正式的历史评价，引用类似古代人物的事迹来类比AI发展中的事件。

需要确保时间线正确，涵盖重要里程碑：图灵测试、达特茅斯会议、AI寒冬、神经网络复兴、深度学习、AlphaGo、GPT等。用古文表达这些事件，可能需要创造一些古式术语，如"算法"称为"算策"，"神经网络"称为"神络"等。

还要注意结构，如开篇的"人工智能本纪"，中间的人物列传，最后的太史公评述。可能需要分段落，每段讲述不同时期或人物，保持史书的严肃性。

最后，检查是否符合《史记》的用词和句式，避免现代用语，同时确保内容准确。例如，用"夫"开头发议论，引用人物的话语，如"辛顿尝言：'神络之妙，在乎深也。'"

现在开始组织内容，先列大纲：本纪部分概述AI发展历程，列传分别描述图灵、达特茅斯群体、辛顿等人物，世家可能涉及重要机构或会议，书部分讨论技术如深度学习，表记录时间线。结尾用太史公曰总结评价。

需要确保各部分衔接自然，符合历史记载的笔法，同时突出关键转折点，如AI寒冬后的复兴，深度学习的突破等。使用古汉语词汇，适当加入虚拟对话或言论，增强真实感。

可能遇到的挑战是如何将现代科技概念转化为古文表达，同时保持准确性。例如，"机器学习"可以译为"机习之术"，"算法"为"算策"等。需保持一致性，避免混淆。

Application: End-to-End RL with Chain of Thought

【以下为《史记》风格的人工智能简史】

人工智能本纪

人工智能者，起于西历一九五六年。达特茅斯会盟，麦卡锡、明斯基、香农等十贤者，共立"Artificial Intelligence"之名。是时也，符号为骨，逻辑为血，欲铸铜铁为圣贤。然寒冬三至，世人皆讥曰："此屠龙之术耳！"

图灵世家

艾伦·图灵，剑桥人也。尝作《机器能思否》，设"模仿之戏"，以辨人机。后世谓之"图灵鉴"。时有司拘其私德，逼饮氰化钾而亡。太史公曰："天若假年，焉知铜丸不可生魂魄耶？"

神经网络列传

罗森布拉特者，造感知机，仿人脑神络。然明斯基作《感知机》书证其弊，遂遭弃市。后有辛顿、勒存、本吉奥三杰，隐于蒙特利尔，三十年磨剑。至乙酉年（2006），倡"深度学习"说，世人方悟神络之妙。辛顿尝言："神络之妙，在乎深也。"

阿尔法狗世家

谷歌有犬，名曰阿尔法。丙申岁（2016），弈韩国李世石于首尔。初局败，后五局四胜，至末子落定，观者皆骇。有棋士柯洁见之，泣曰："此非人力可及也！"犬竟得九段衔，棋道自此改弦更张。

大语言模型书

OpenAI者，硅谷异士聚所也。庚子岁（2020），作GPT-3，识古今典籍八百万卷。其言煌煌如圣训，其思缈缈若列子御风。微软注资百亿金，纳其为肱骨。然有识者忧："此物善谀，恐乱人间正声。"

太史公曰

观AI七十年，犹见扁鹊三见桓侯：初视腠理，再窥肌肤，终入骨髓。昔庄子疑濠梁鱼乐，今机器竟解人情。然则图灵鉴中，铜铁或具七窍心，而七窍心者，渐似铜铁矣！悲夫！

Application: End-to-End RL with Chain of Thought

请用滕王阁序的风格来一段人工智能简史

Thought for 17 seconds

好的，用户之前让我用鲁迅和梁启超的风格写了人工智能简史，现在又要求用《滕王阁序》的风格来写。我需要先回忆一下《滕王阁序》的特点，比如华丽的辞藻、对仗工整、用典丰富、结构严谨，以及那种壮阔的意境。

首先，得确定滕王阁序的结构，通常开头会描绘场景，接着叙述历史或事件，然后转入抒情或议论。我需要将人工智能的发展历程融入这样的结构中。

接下来，要找到合适的典故和比喻，比如用古代的名人、地名来对应AI发展中的关键人物和事件。例如，图灵可以比作古代的先贤，达特茅斯会议可以比作盛会，寒冬期可以用自然现象来隐喻。

然后，要注意对仗和四六骈文的句式，比如"星分翼轸，地接衡庐"这样的结构。需要确保每句的字数和结构对称，同时押韵，保持古典韵味。

还要考虑用户可能的深层需求，他们可能希望这段简史不仅有文学性，还要涵盖AI的重要里程碑，如图灵测试、达特茅斯会议、神经网络的发展、深度学习的突破，以及现代应用如ChatGPT等。

需要检查是否有遗漏的重要事件，并确保时间线合理。同时，避免过于晦涩的用典，让读者即使不熟悉《滕王阁序》也能大致理解内容。

最后，结尾部分要模仿滕王阁序的抒情，表达对未来的展望和感慨，呼应原文中的"呜呼"或"胜地不常"等结构。确保整体气势恢宏，既有历史厚重感，又有科技的未来感。

Application: End-to-End RL with Chain of Thought

【以下为《滕王阁序》风格的人工智能简史】

时维庚子，序属寒秋。观硅谷之烟霞，访智械之灵枢。图灵铸鉴，光照英伦三岛；麦卡锡立言，气吞达特茅斯。逻辑符号，列如北辰悬天；感知雏形，灿若南溟蜃楼。然寒冬骤至，霜凝明斯基之镜；长夜未央，雾锁西蒙之舟。

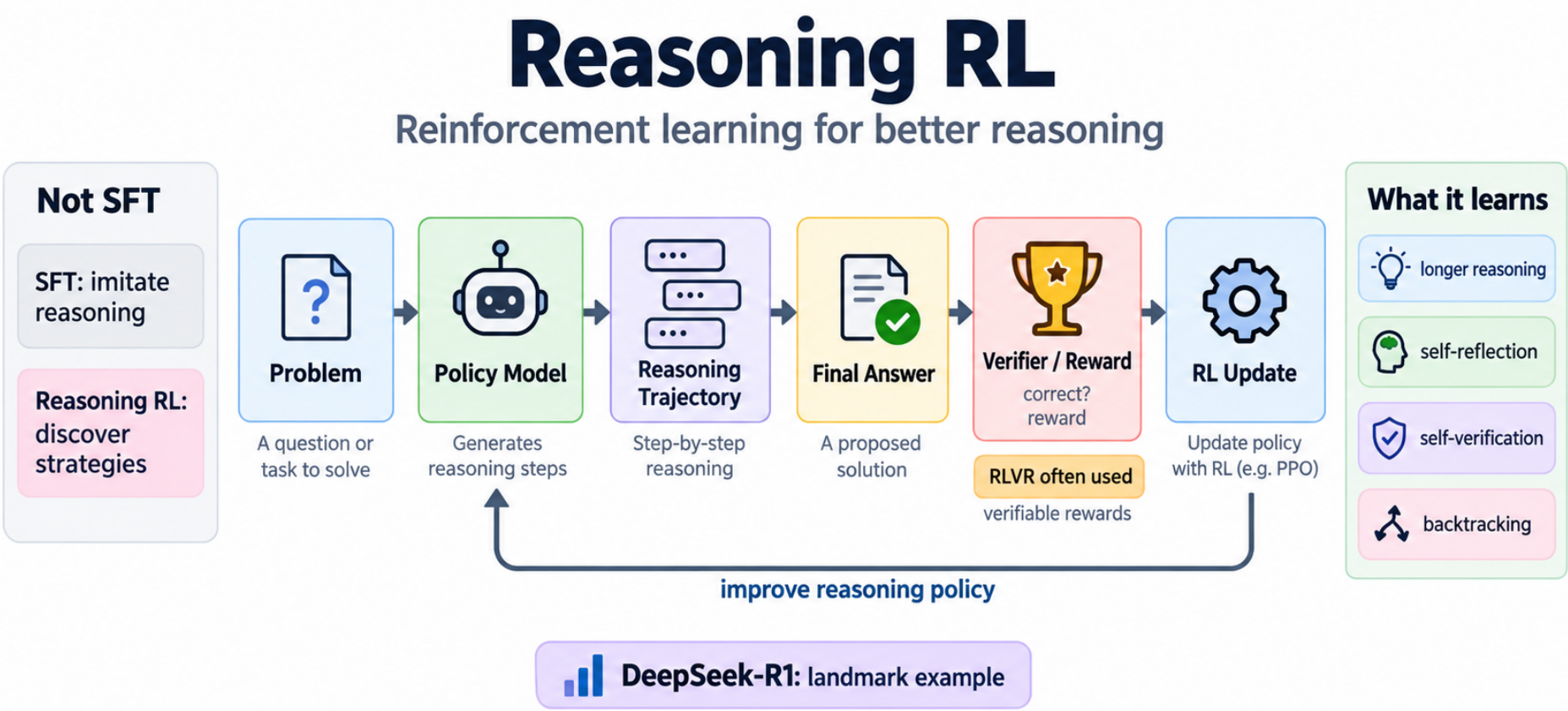
幸有罗森布拉特，临危受命，织神经网络如绣云锦；辛顿独行，潜修深度学习似琢昆玉。蒙特利尔地，三十载寒窗悟道；多伦多城，一朝间雷破九霄。卷积识万象，犹持轩辕之鉴；循环通古今，恍闻河洛之书。

至若阿尔法狗弈破星阵，生成对抗画染丹青。transformer执牛耳，吞吐八荒文字；大模型展鹏翼，扶摇九天风云。GPT振铎，声震稷下学宫；BERT燃犀，光耀鸿都门观。此皆硅基文明之盛筵，实乃碳基智慧之遗响。

呜呼！寒暑易节，终有算力破茧之日；春秋代序，岂无代码化蝶之时？冯诺依曼已逝，香农公何在？望硅晶之原野，烟光凝而算符紫；临数据之江渚，比特涌而信息流。天工开物，当惊寰宇殊异；后之览者，亦将有感于斯文！

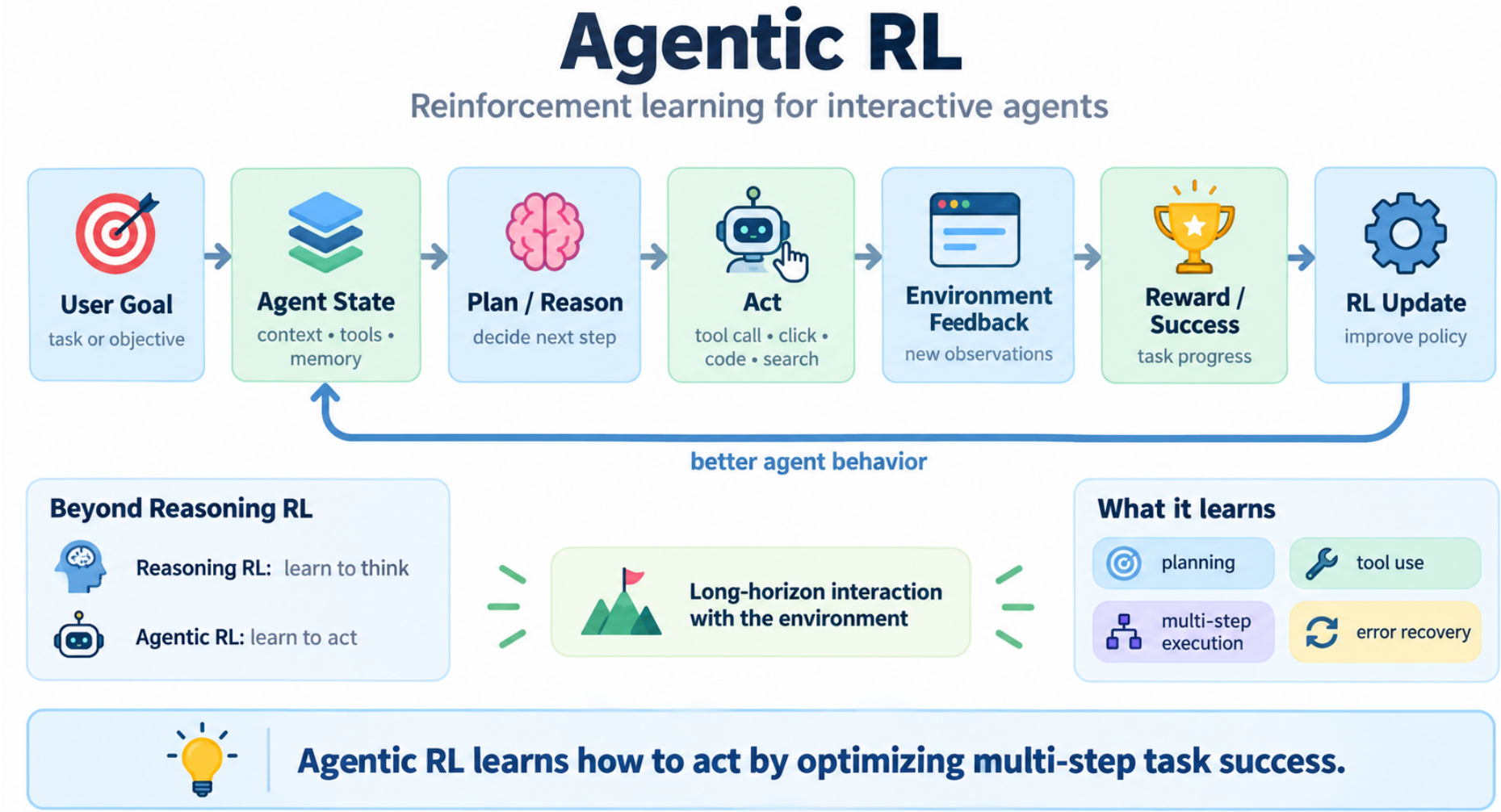
Recent Successes in RL: Reasoning RL

Recent Successes in RL: Reasoning RL

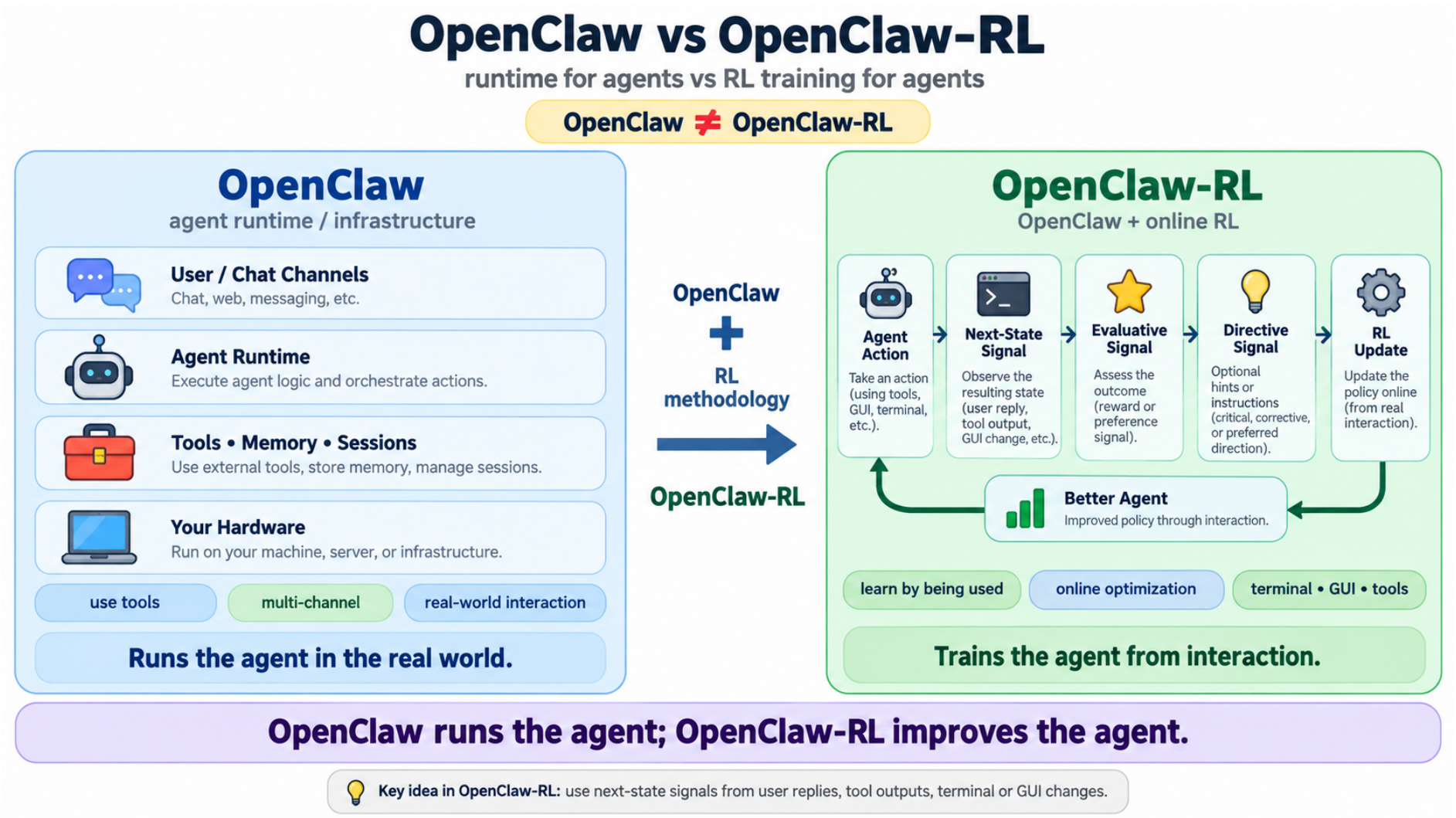


Reasoning RL learns how to reason by optimizing for successful problem solving.

Recent Successes in RL: Agentic RL



Recent Successes in RL: Agentic RL



Recent Successes in RL: Agentic RL

Outline

1. Intelligence & RL
2. People: TA, Professor and His Academic Family
3. Course Information
4. Conclusion

TA

- 张天翼
- 郑美涵

Professor Ziyu Shao: 邵子瑜

- 领导 IID 实验室

https://faculty.sist.shanghaitech.edu.cn/faculty/shaozy/home.html

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Course Features

- Solid Foundation of RL: modeling analysis and understanding of theoretical proof
- Computational Thinking: algorithmic perspective and programming practice
- Application Mentality: many examples from AI and related applications
- Research-Oriented: aims to produce high-quality original research results and publish in top conferences & journals

Time & Venue

- Time: Every Tuesday and Thursday 8:15-9:55am
- Location: Room 103, Teaching Center
- Course Forum in Piazza: `https://piazza.com/shanghaitech.edu.cn/fall2026/si252/`

Course Grade

- Homework: 20% (10 problem sets).
- Final Exam: 50% (12th week).
- Final Project: 25%.
- Course Behavior: 5% .

Homework Policy

- Write the homework with LaTeX and output it as a PDF file.
- Write the programming assignment with Python
- **Later** homework receives no credit.
- You are allowed to discuss with others and use any references, but if you do so please list your collaborators and cite your references for each question.
- Not writing your own solutions or not listing your collaborators or not citing your references may be considered plagiarism.

Project Policy

- Project can be done by a team with ≤ 3 students.
- Teams with four and more students are NOT allowed.
- Your team is required to use Python for the programming part.
- Your team needs to submit the iPython Notebook including your source codes, simulation results, analysis, discussions, tables, figures, etc.
- Your team also needs to submit the final report using any LaTeX template
- **Later** Project receives no credit.

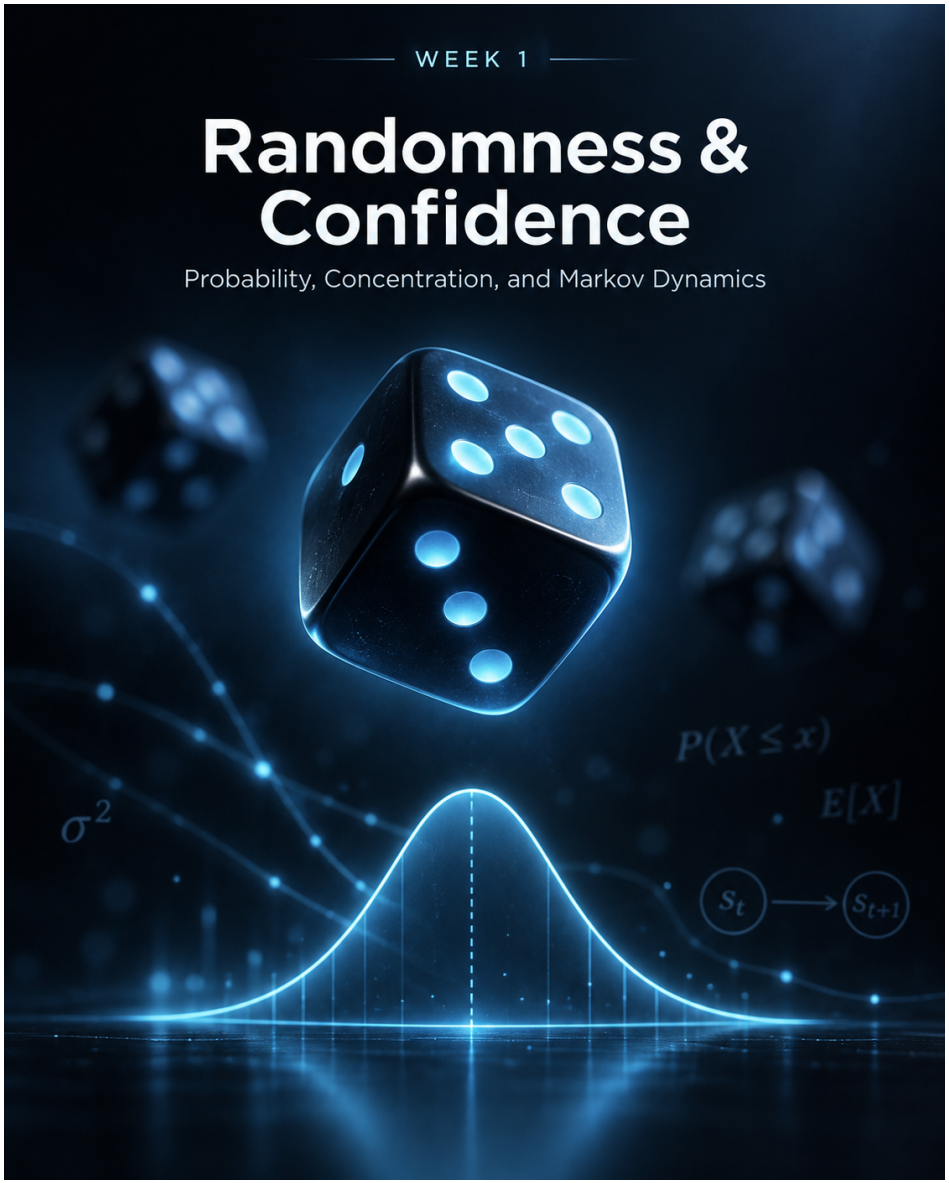
LaTeX Setup

- LaTeX Package (Unix, Windows): TeXLive (*https://tug.org/texlive/*)
- LaTeX Package (MacOs): MacTeX (*http://tug.org/mactex*)
- Popular LaTex Editors
  - Texpad (MacOs): *https://www.texpad.com*
  - TeXstudio (Windows,MacOs): *http://www.texstudio.org*
  - LyX(Windows,MacOs): *https://www.lyx.org*
  - WinEdt (Windows): *http://www.winedt.com/*

LaTeX Tutorials

- Wiki LaTeX Book: *http://en.wikibooks.org/wiki/LaTeX*
- Collection of links for LaTeX: *https://www.tug.org/begin.html*
- Forum: *https://tex.stackexchange.com/*

Course Schedule (Tentative)



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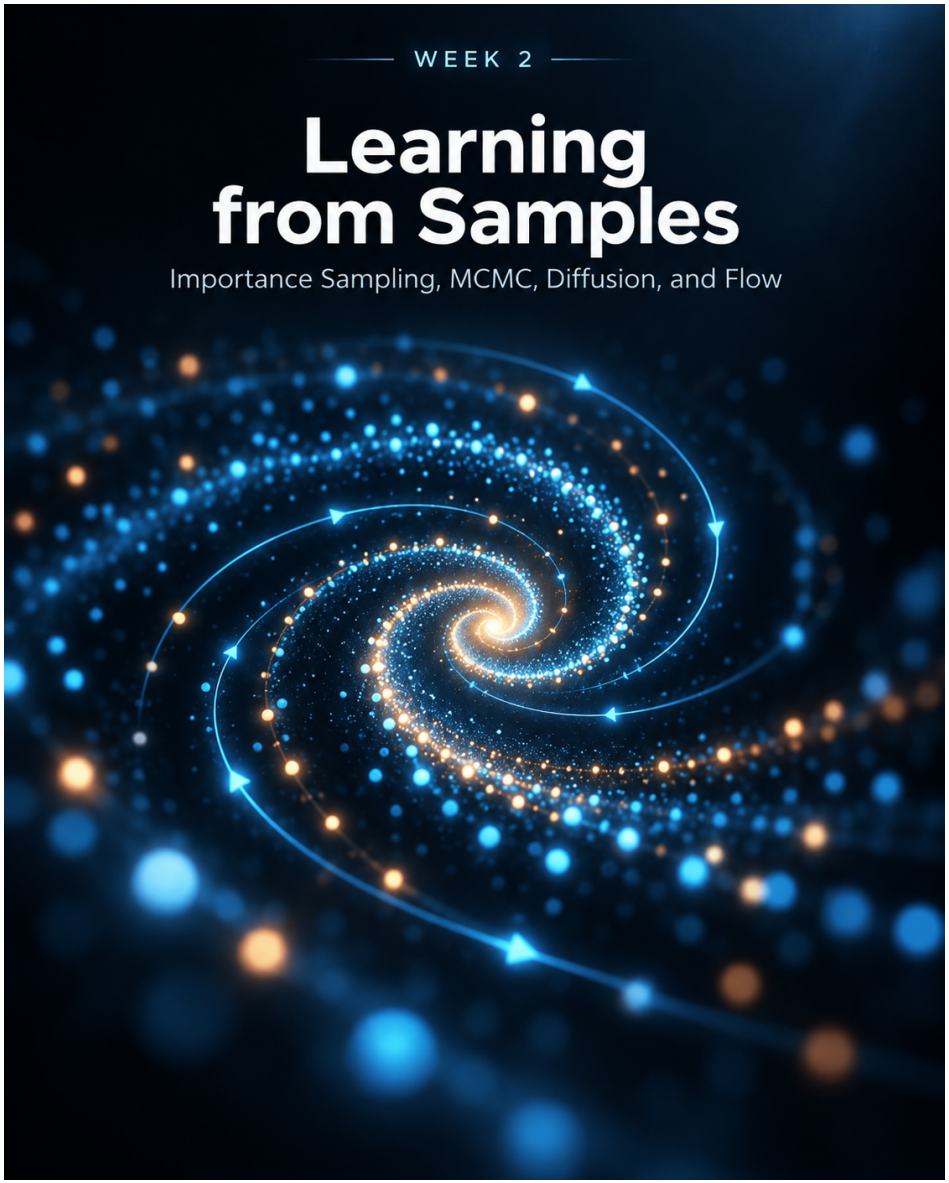

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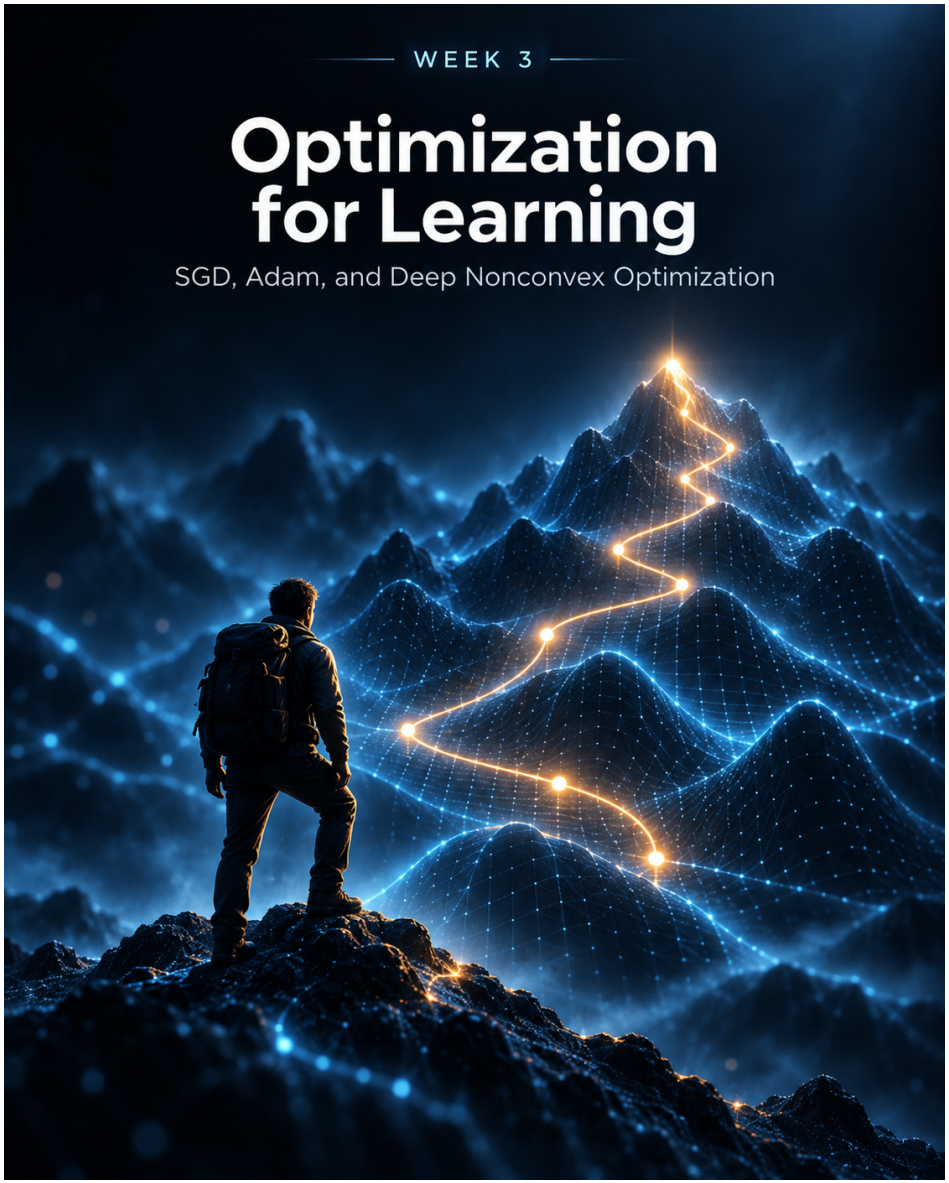

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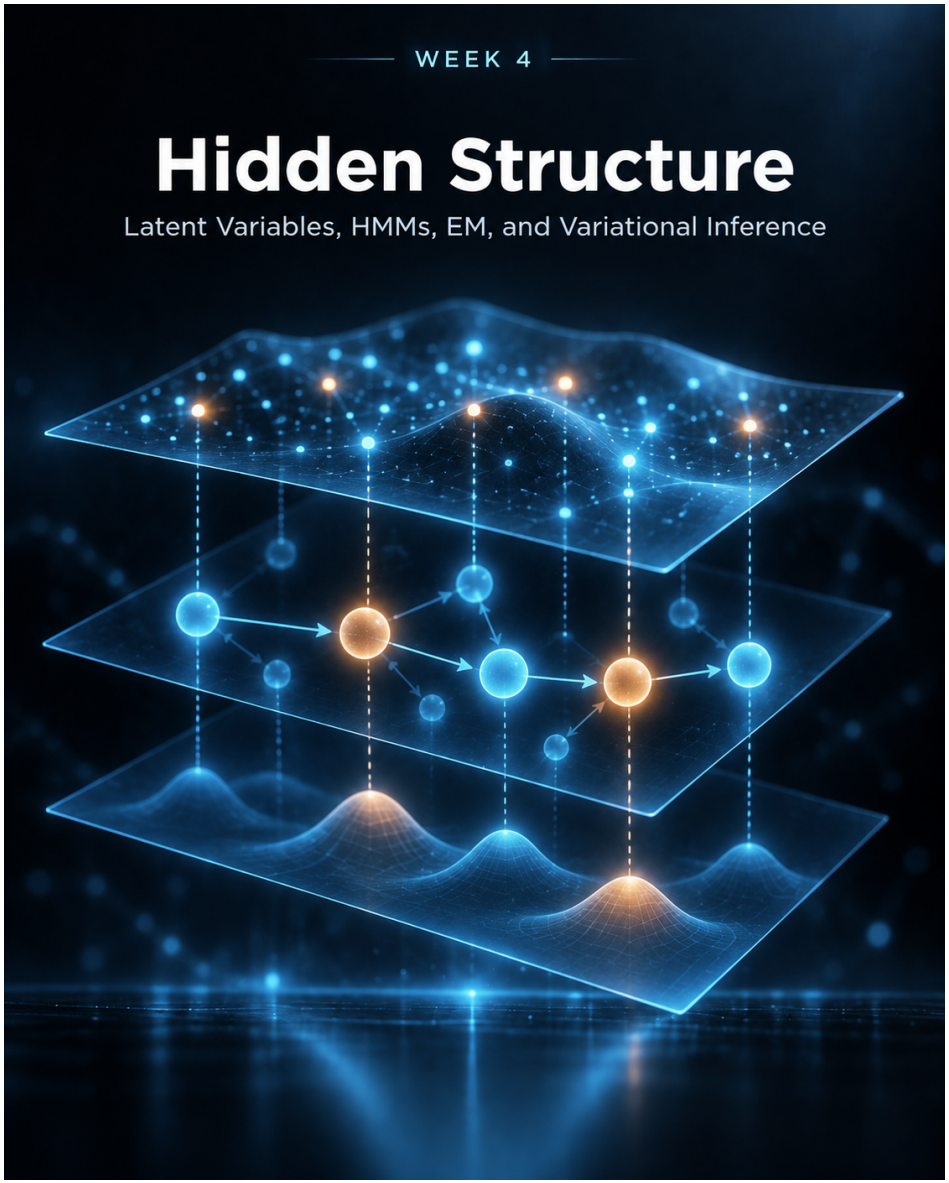

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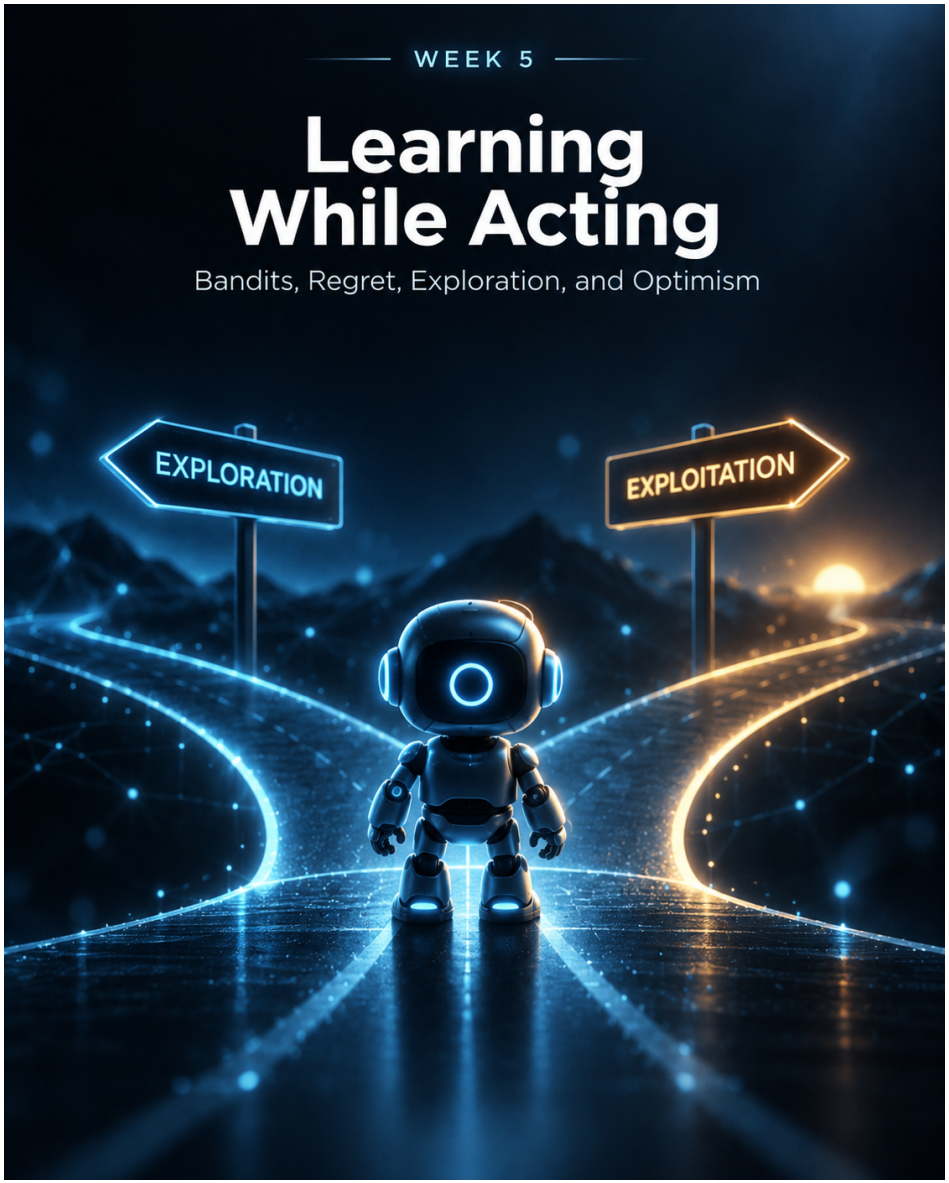

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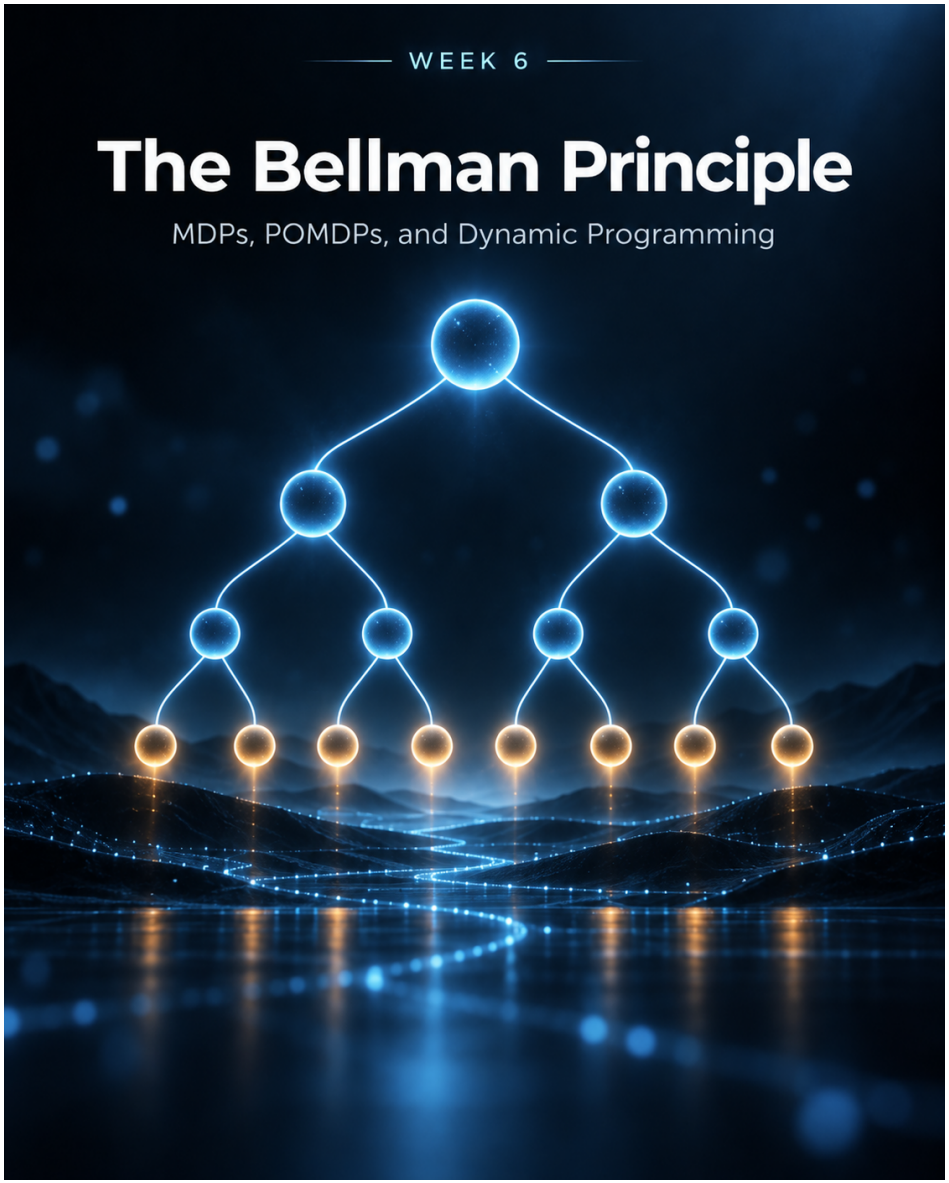

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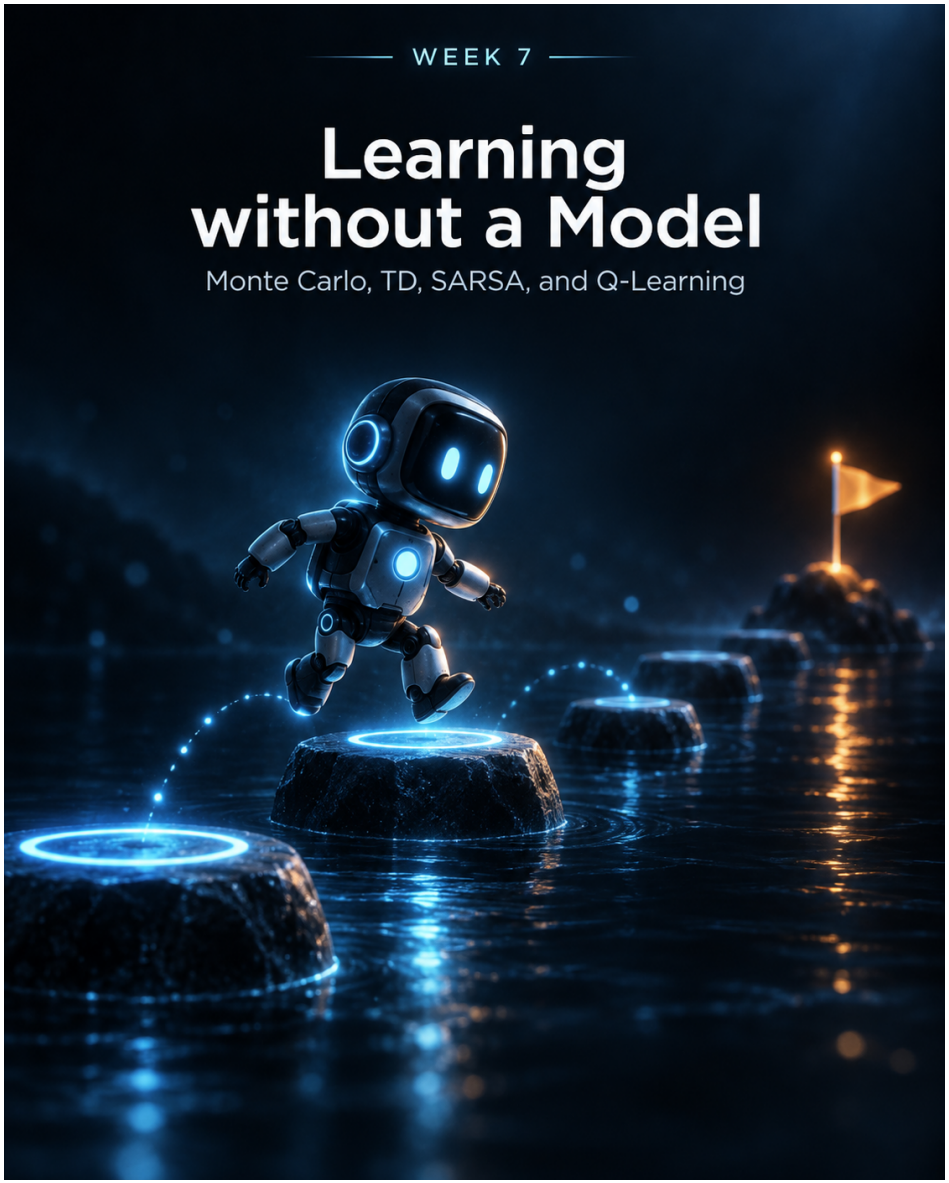

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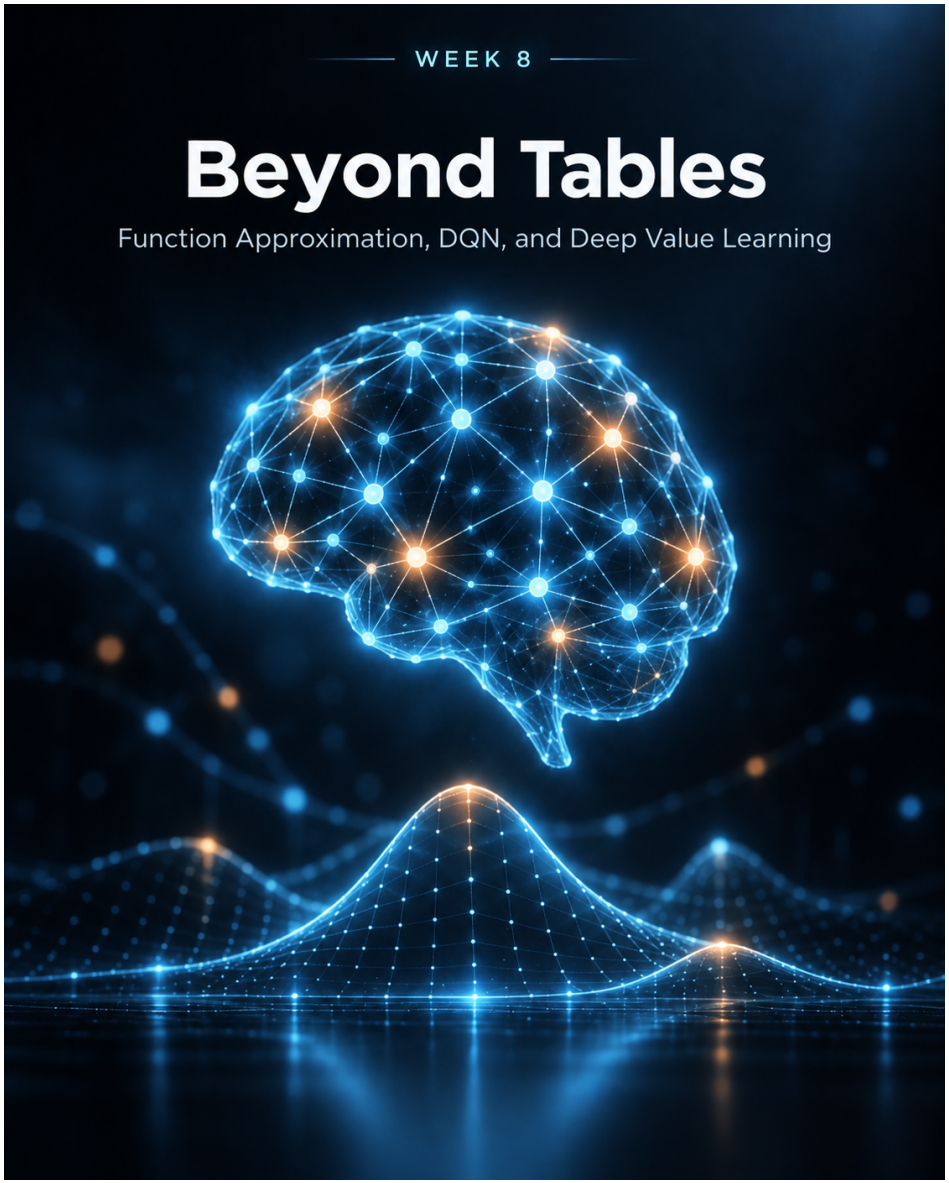

Course Schedule (Tentative)

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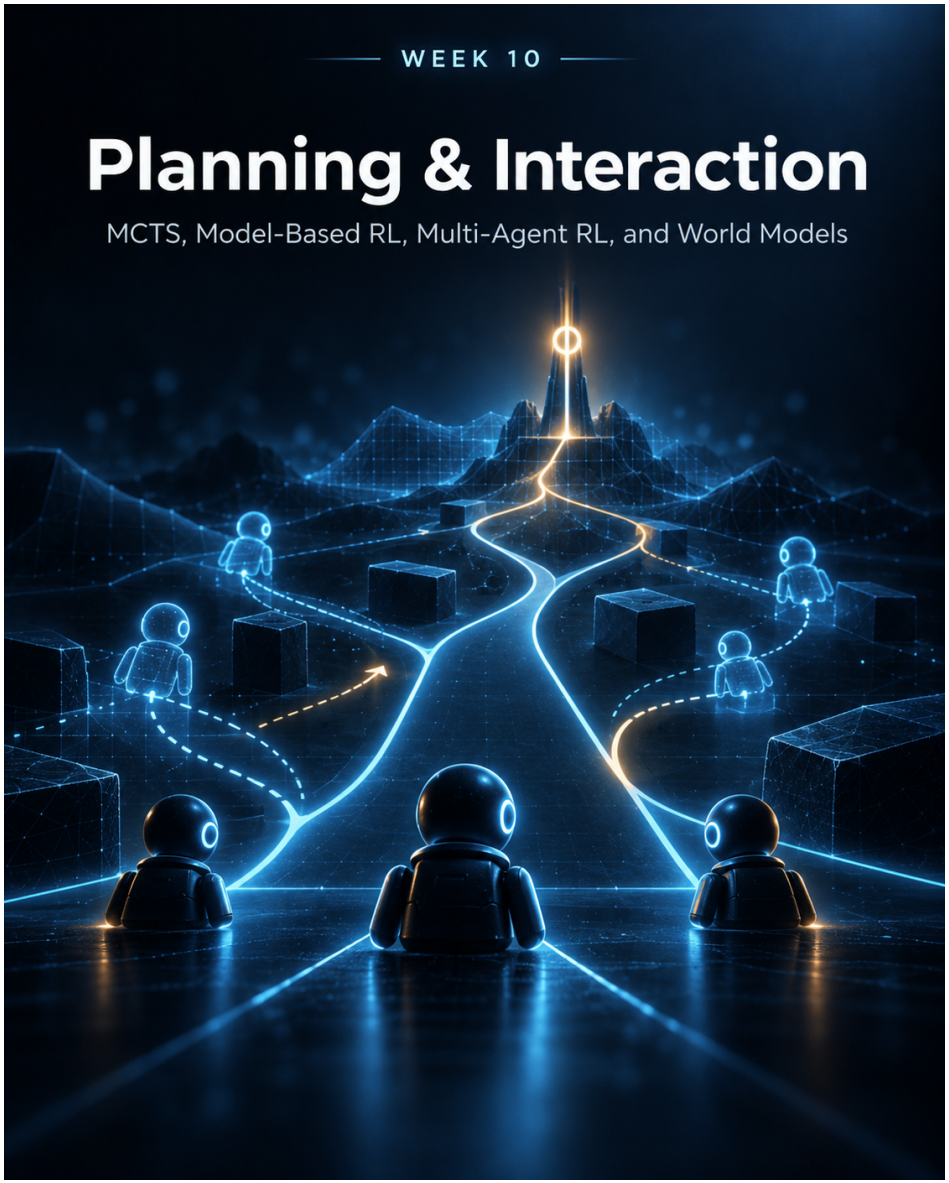

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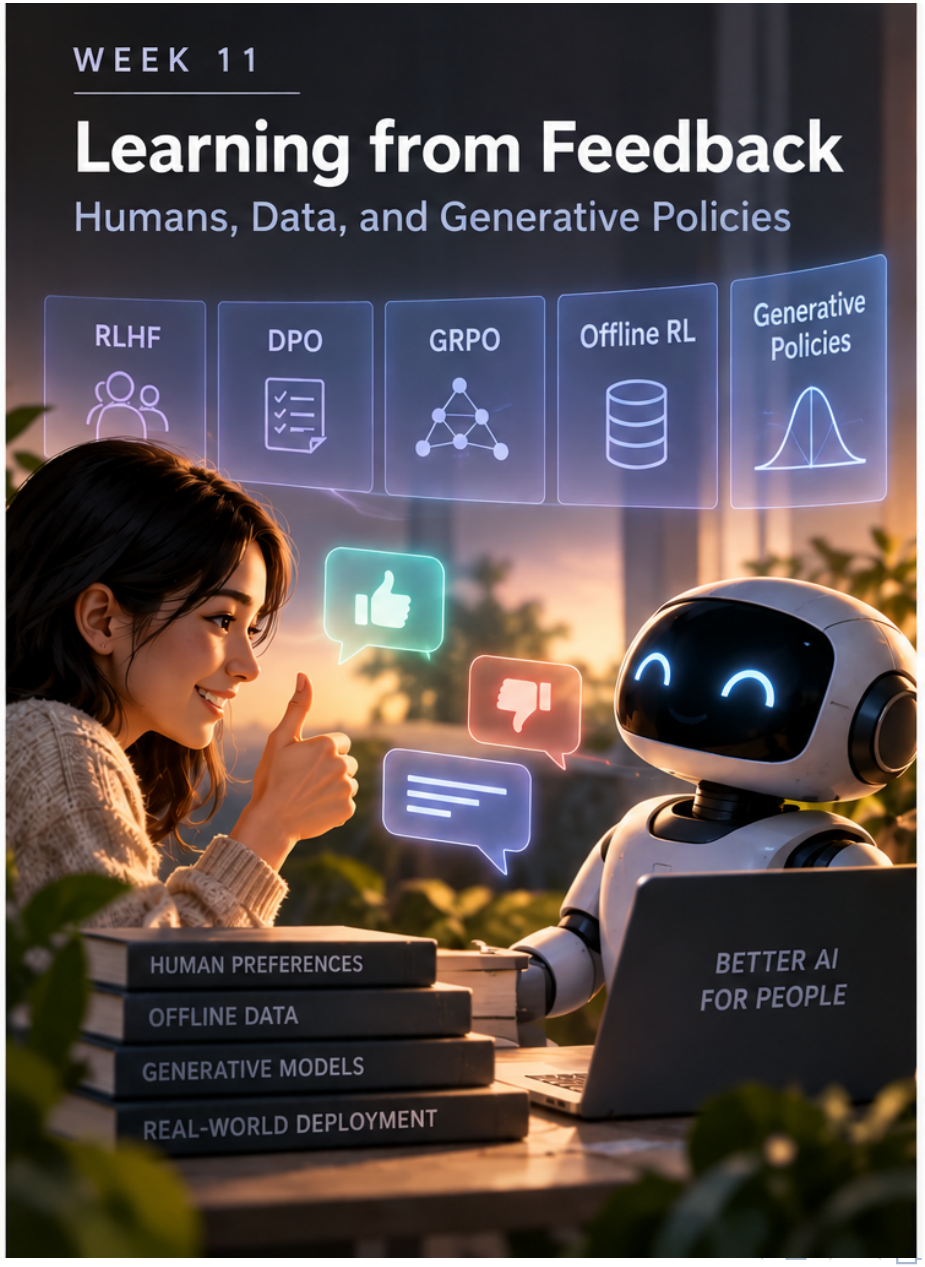

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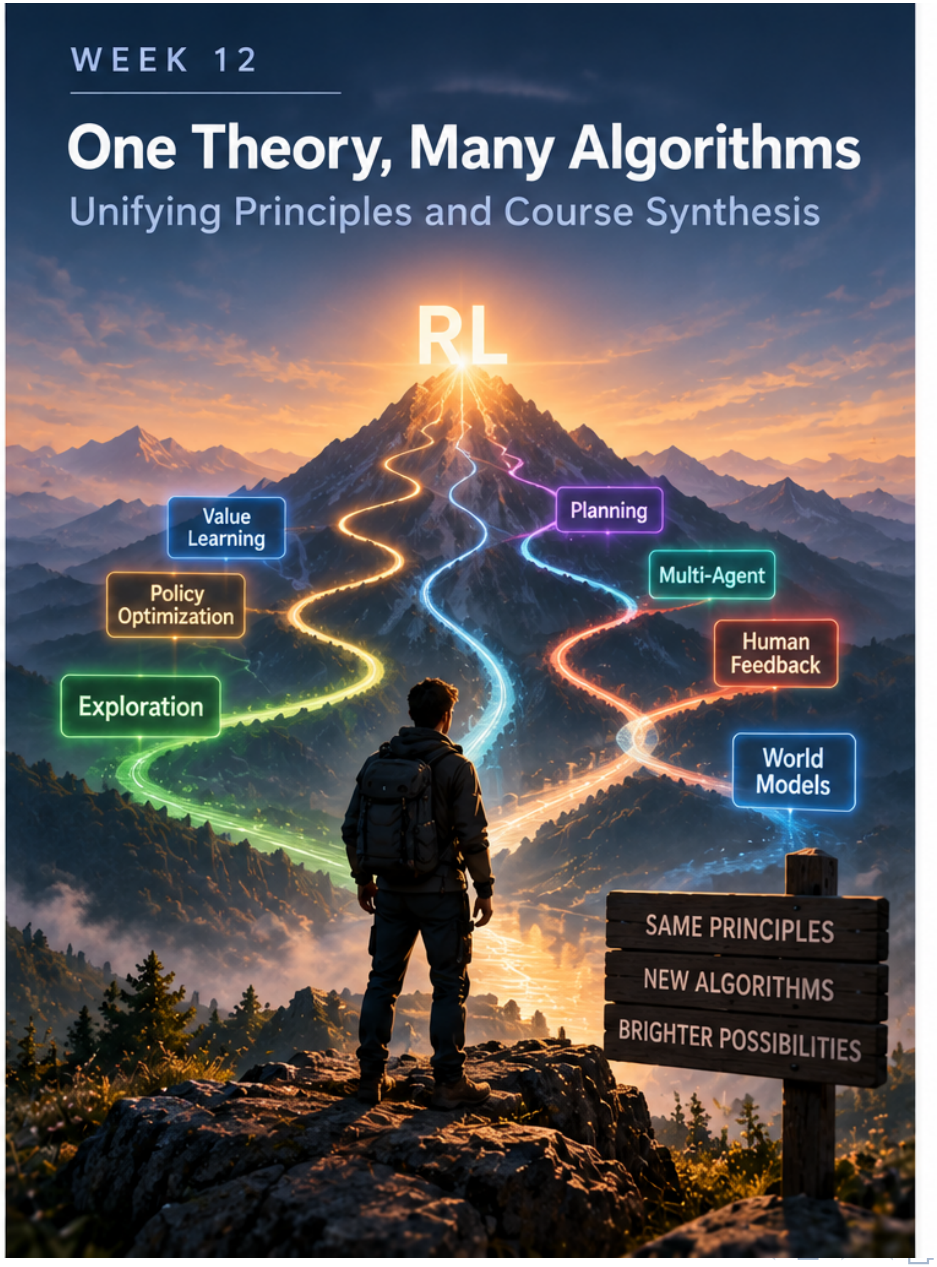
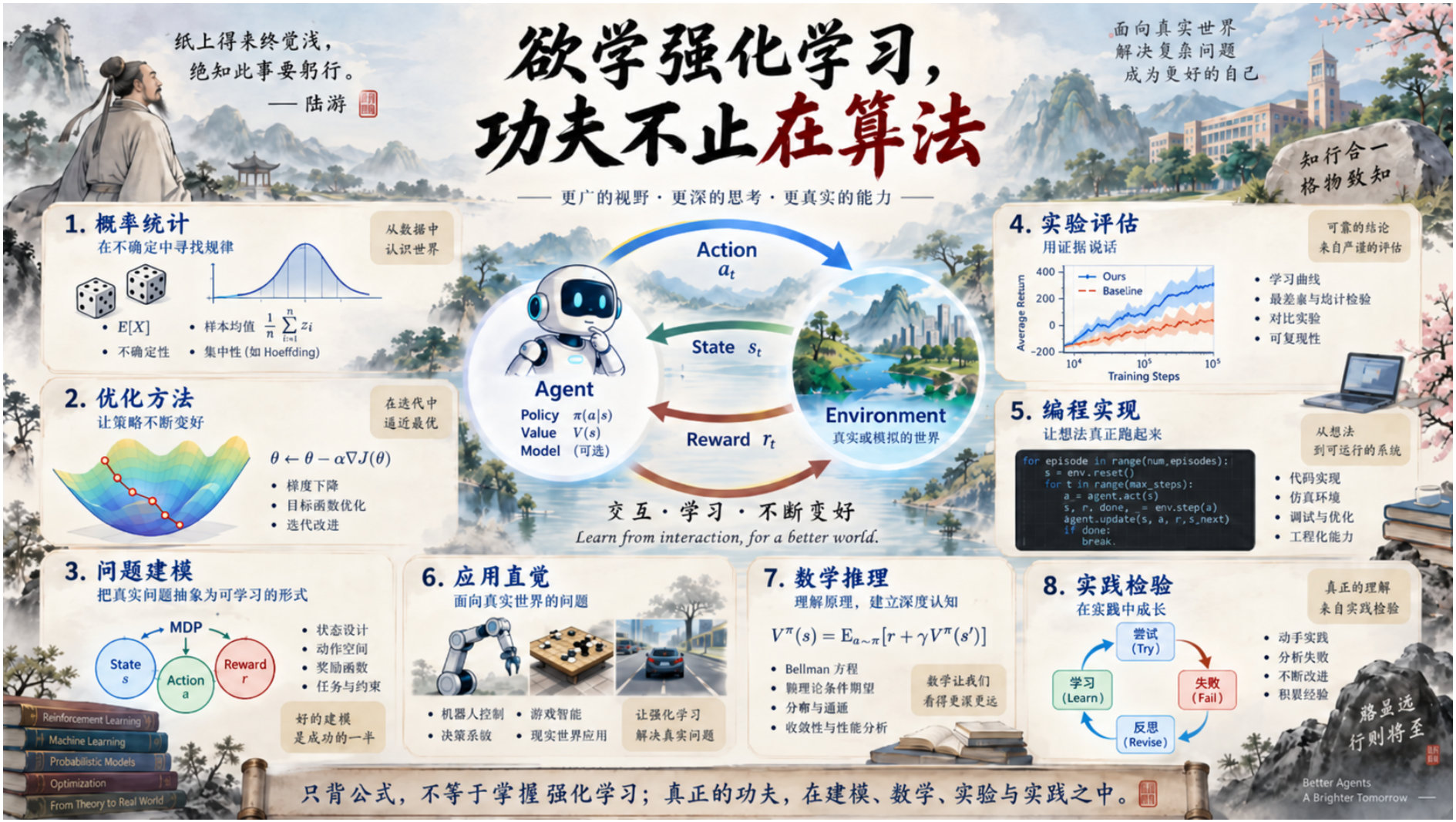


Textbooks

John R. Birge
François Louveaux
Introduction to Stochastic Programming
Second Edition
Michael C. Fu Editor
Handbook of Simulation Optimization
Princeton Series in APPLIED MATHEMATICS
Robust Optimization
Introduction to Decision Analysis
A Practitioner's Guide to Improving Decision Quality
SECOND EDITION
Approximate Dynamic Programming
Solving the Curses of Dimensionality
Warren B. Powell
Optimal Learning
MULTI-ARMED BANDIT ALLOCATION INDICES
SECOND EDITION
Dynamic Programming and Optimal Control
APPROXIMATE DYNAMIC PROGRAMMING
Dimitri P. Bertsekas
SECOND EDITION
Model Predictive Control
OPTIMAL CONTROL
Frank L. Lewis
INTRODUCTION TO STOCHASTIC SEARCH AND OPTIMIZATION
Estimation, Simulation, and Control
JAMES C. SPALL
MORGAN & CLAYPOOL PUBLISHERS
Active Learning
Burr Settles
WILEY
43
Jiongmin Yong
Xun Yu Zhou
Stochastic Controls
Hamiltonian Systems and HJB Equations
Reinforcement Learning
An Introduction
second edition
Richard S. Sutton and Andrew G. Barto
Markov Decision Processes
Discrete Stochastic Dynamic Programming
MARTIN L. PUTERMAN
Online Computation and Competitive Analysis
Allan Borodin
Ran El-Yaniv
STOCHASTIC SIMULATION OPTIMIZATION
An Optimal Computing Budget Allocation
Chun-Hung Chen • Loo Hay Lee
Slide 28

汝果欲学诗，
功夫在诗外。
陆游
游
学诗不止于辞章，更在学问、阅历、见识与胸怀。
博览群书
读万卷书
开阔眼界
庚癸全书
史记
体察人生
观人间百态
知世事冷暖
亲近山水
观山川草木
感自然之美
纸上得来终觉浅
绝知此事要躬行
——陆游
感受四时
春花秋月冬雪
体会生命流转
行走四方
见不同风土
增广见识
行万里路
诗在心中
而不止于纸上
心系家国
关怀社会天下
涵养胸怀
读诗心
连着大千世界


纸上得来终觉浅，
绝知此事要躬行。
—— 陆游
欲学强化学习，
功夫不止在算法
—— 更广的视野 · 更深的思考 · 更真实的能力 ——
面向真实世界
解决复杂问题
成为更好的自己
知行合一
格物致知
1. 概率统计
在不确定中寻找规律
从数据中
认识世界
$E[X]$
样本均值 $\frac{1}{n}\sum_{i=1}^{n} z_i$
不确定性
集中性 (如 Hoeffding)
2. 优化方法
让策略不断变好
在迭代中
逼近最优
$\theta \leftarrow \theta - \alpha \nabla J(\theta)$
梯度下降
目标函数优化
迭代改进
Action
a_t
State s_t
Reward r_t
Agent
Policy $\pi(a|s)$
Value $V(s)$
Model (可选)
Environment
真实或模拟的世界
交互 · 学习 · 不断变好
Learn from interaction, for a better world.
4. 实验评估
用证据说话
可靠的结论
来自严谨的评估
Ours
Baseline
Average Return
Training Steps
学习曲线
最差值与均计检验
对比实验
可复现性
5. 编程实现
让想法真正跑起来
从想法
到可运行的系统
for episode in range(num_episodes):
s = env.reset()
for t in range(max_steps):
a = agent.act(s)
s, r, done, _ = env.step(a)
agent.update(s, a, r, s_next)
if done:
break
代码实现
仿真环境
调试与优化
工程化能力
3. 问题建模
把真实问题抽象为可学习的形式
MDP
State
Action
Reward
状态设计
动作空间
奖励函数
任务与约束
好的建模
是成功的一半
Reinforcement Learning
Machine Learning
Probabilistic Models
Optimization
From Theory to Real World
6. 应用直觉
面向真实世界的问题
机器人控制
游戏智能
决策系统
现实世界应用
让强化学习
解决真实问题
7. 数学推理
理解原理，建立深度认知
$V^{\pi}(s) = \mathrm{E}_{a\sim\pi}[r + \gamma V^{\pi}(s')]$
Bellman 方程
期望条件期望
分布与通道
收敛性与性能分析
数学让我们
看得更深更远
8. 实践检验
在实践中成长
真正的理解
来自实践检验
尝试 (Try)
失败 (Fail)
反思 (Revise)
学习 (Learn)
动手实践
分析失败
不断改进
积累经验
路虽远
行则将至
只背公式，不等于掌握强化学习；真正的功夫，在建模、数学、实验与实践之中。
Better Agents
A Brighter Tomorrow

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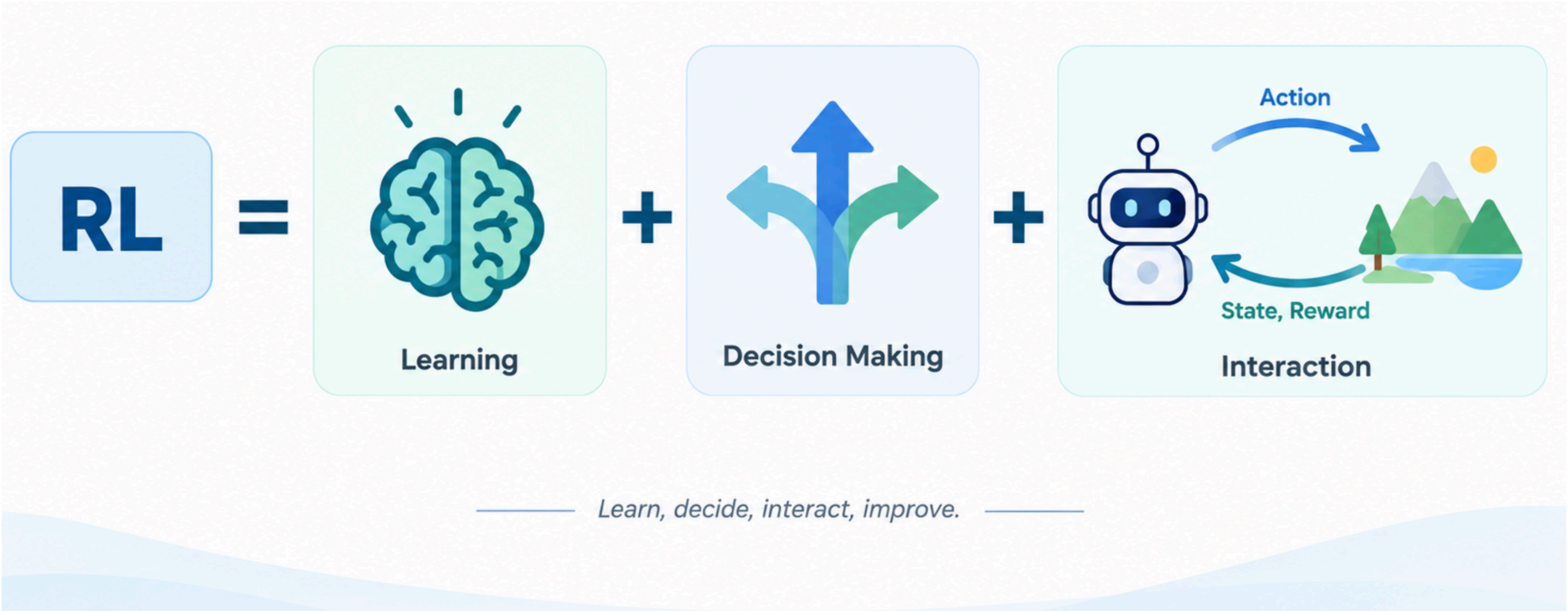
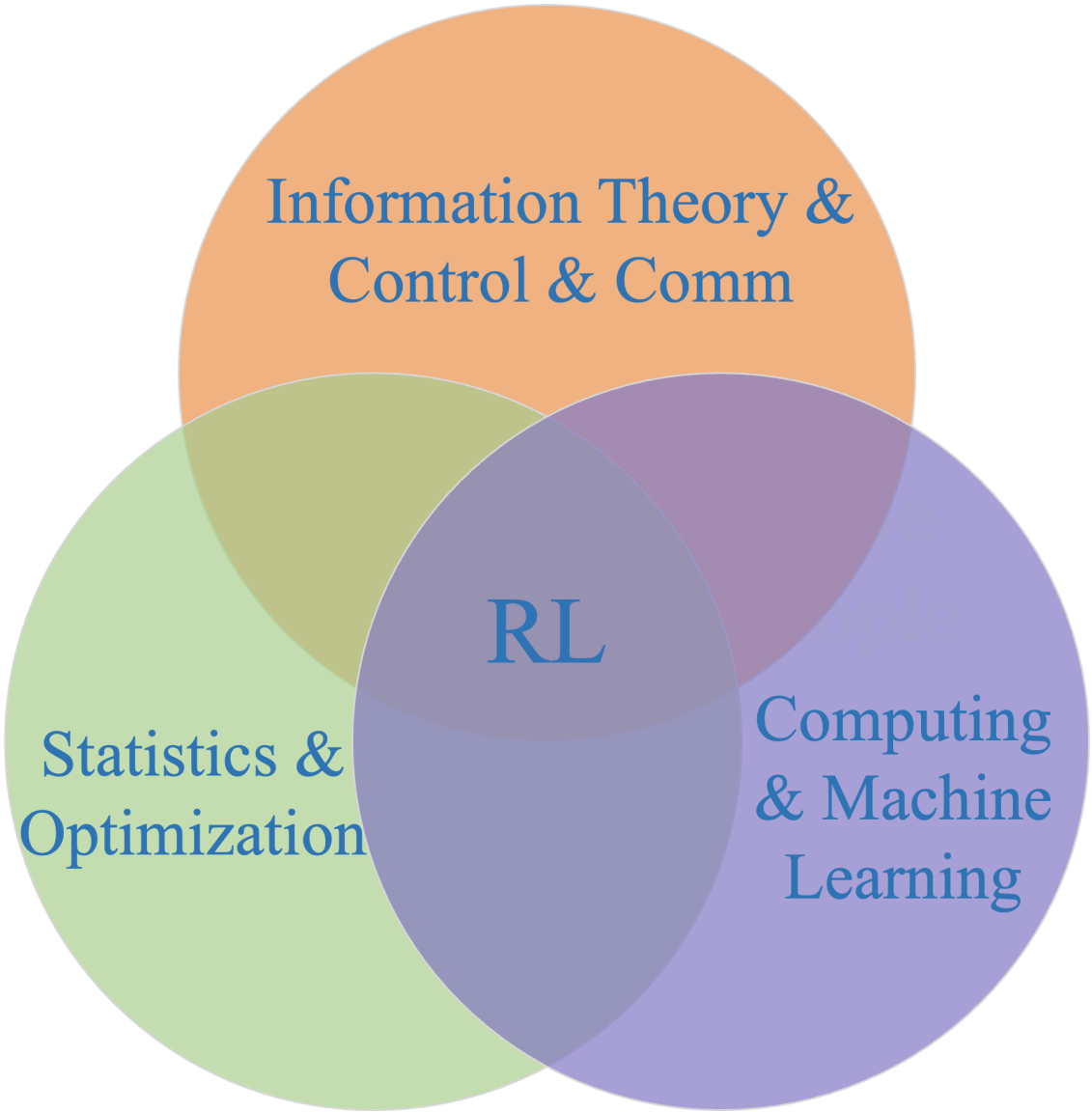
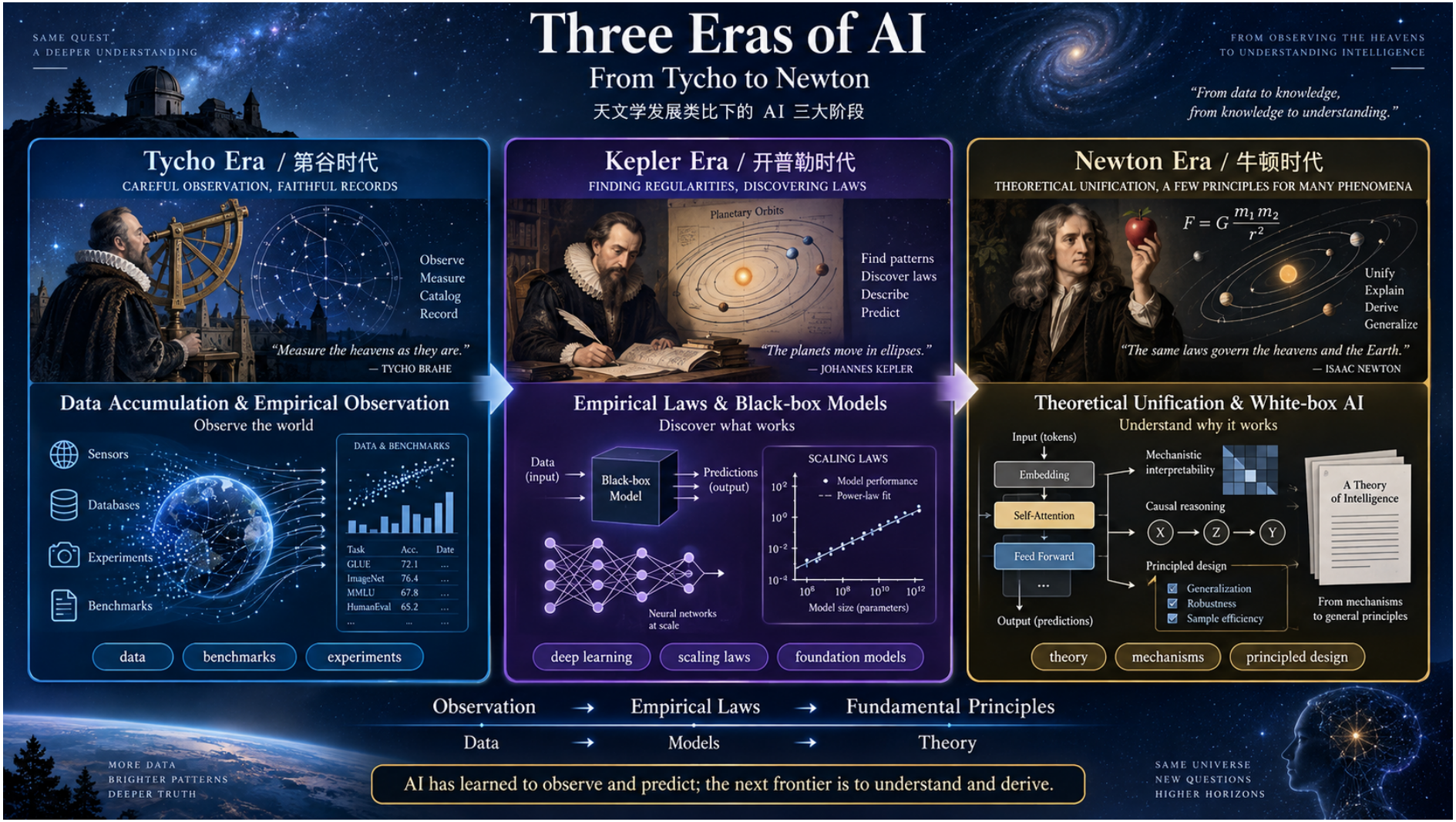
Essence of Reinforcement Learning



Reinforcement Learning: Interdisciplinary Topic

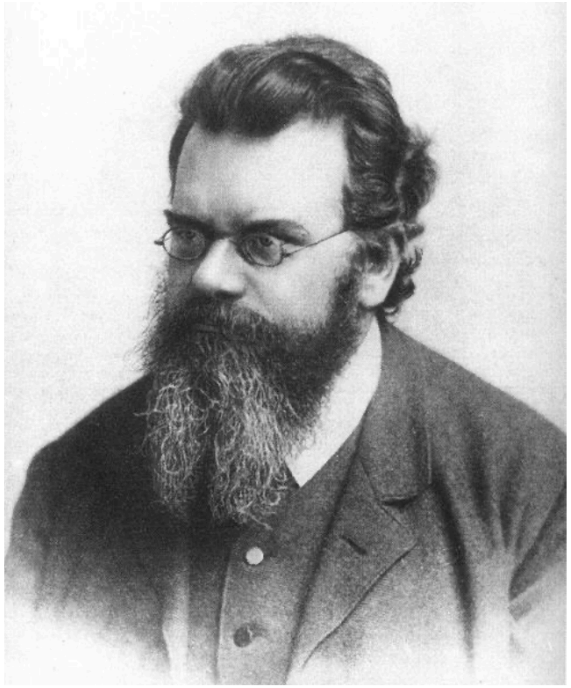


Three Eras of AI: From Art to Science



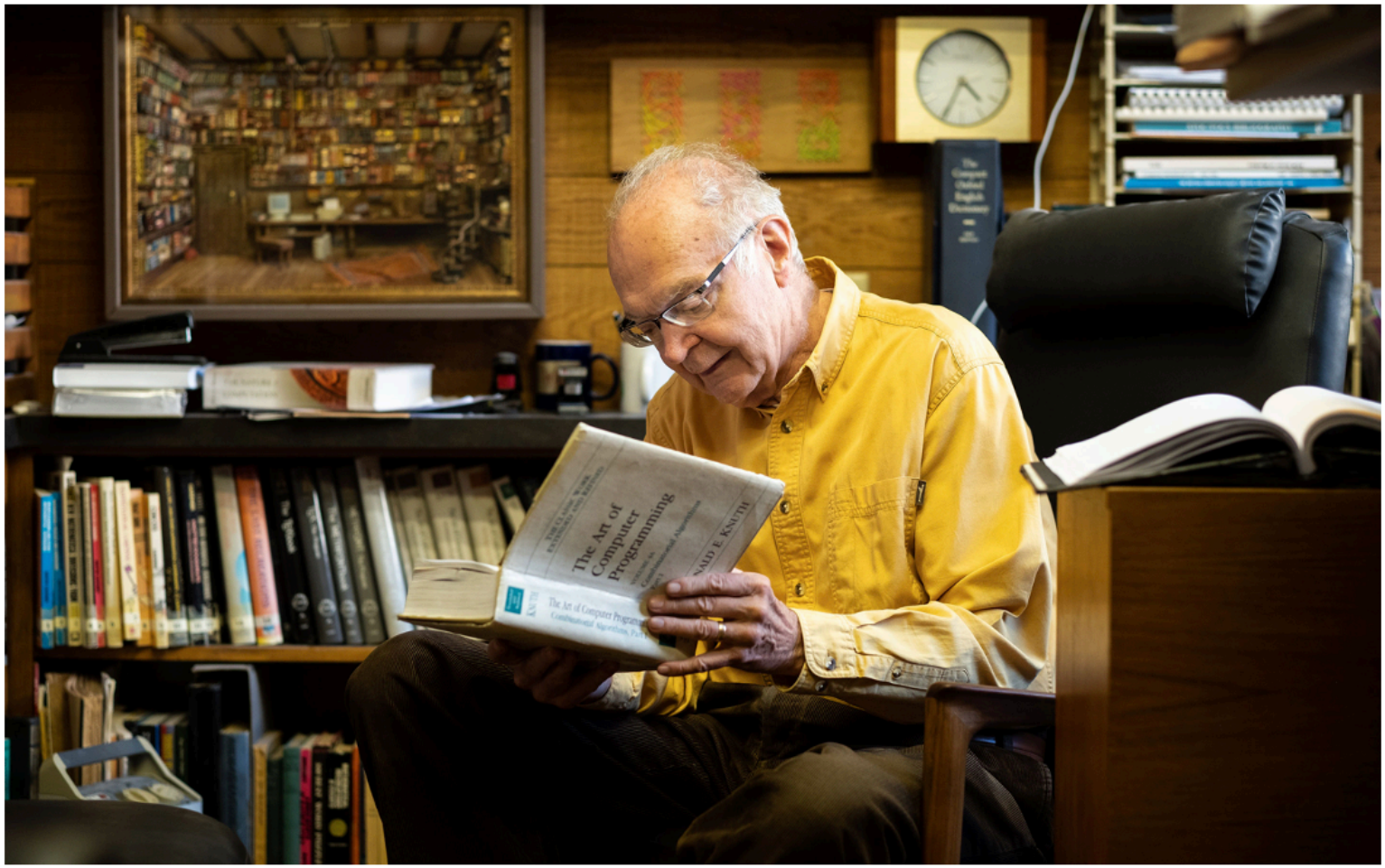
"Nothing is more practical than a good theory! "

Ludwig Boltzmann



"The best theory is inspired by practice,
the best practice is inspired by theory! "

-- Donald E. Knuth



Let the Adventure Begin!